

Descriptive power and predictive limits of a discrete Hasimoto–DNLS model of protein backbone structure

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ABSTRACT

Determining the three-dimensional structure of a protein from its amino-acid sequence remains a fundamental problem in biophysics. The discrete Frenet geometry of the C_α backbone can be mapped, via a Hasimoto-type transform, onto a complex scalar field $\psi = \kappa e^{i\int \tau ds}$ satisfying a discrete nonlinear Schrödinger equation (DNLS), whose soliton solutions reproduce observed secondary-structure motifs. Whether this mapping, which provides a compact geometric description of folded states, can be extended to a predictive framework for protein folding is an open question. We derive an exact closed-form decomposition of the DNLS effective potential $V_{\text{eff}} = V_{\text{re}} + iV_{\text{im}}$ in terms of curvature ratios and torsion angles, validating the result to machine precision across 856 non-redundant proteins. Our analysis identifies three structural barriers to forward prediction: (i) V_{im} encodes chirality through the odd symmetry of $\sin \tau$; its magnitude is $\sim 31\%$ that of the real part, and neglecting it introduces a 2^N degeneracy; (ii) V_{re} is determined primarily ($\sim 95\%$) by local geometry, leaving its explicit sequence dependence below $\sim 5\%$ of the variance; and (iii) self-consistent field iterations fail to recover native structures (mean RMSD = 13.1 Å) even with hydrogen-bond terms, yielding torsion correlations indistinguishable from zero. On the constructive side, we show that the residual of the DNLS dispersion relation serves as a geometric order parameter for α -helices (ROC AUC = 0.72), identifying them as the regions where the backbone most closely approximates an integrable system. These findings indicate that the Hasimoto map functions as a kinematic identity rather than a dynamical governing equation. The obstacles to *ab initio* prediction arise from the purely local, real-potential reduction built upon it, not from the lossless map itself.

1. Introduction

The relationship between amino-acid sequence and three-dimensional structure is a central problem in molecular biophysics. One geometric approach treats the backbone as a one-dimensional filament whose shape is captured by its curvature and torsion. This idea originates in fluid mechanics: Hasimoto [1] showed that the local induction approximation for a vortex filament can be exactly transformed, via the complex scalar field $\psi = \kappa e^{i\int \tau ds}$, into the cubic nonlinear Schrödinger equation (NLS). Because the NLS is completely integrable, this transformation converts the geometric evolution of a three-dimensional curve into a one-dimensional soliton problem with exact analytical solutions.

An analogous construction applies to the protein backbone, with one difference: the curve is discrete. It is represented as a sequence of points in $\mathbb{R}^3$ whose local shape is characterized by two scalar fields, the bond angle $\kappa[n]$ and the torsion angle $\tau[n]$ at each C_α vertex; together these

two fields determine the conformation up to rigid-body motion. A discrete Hasimoto-type transform then encodes these two fields as a single complex scalar field, and the resulting geometric language has proven highly successful at *describing* native conformations. Since the native fold is ultimately encoded in the amino-acid sequence [2], it is natural to ask whether a *local, integrable reduction* of the backbone geometry can do more than describe, that is, whether it can *generate* the three-dimensional fold from sequence-level input alone. This question is distinct and largely uncharacterized.

Niemi and collaborators pursued this analogy systematically for protein backbones, developing a geometric program rooted in gauge field theory and nonlinear dynamics [3–6]. Working with the discrete Frenet frame of the C_α chain, they constructed a generalized discrete nonlinear Schrödinger equation (DNLS) whose dark-soliton solutions reproduce the characteristic (κ, τ) profiles of α -helices and β -strands. Molkenhuth, Hu, and Niemi [7] showed that a two-soliton configuration reproduces the villin headpiece HP35 with RMSD = 0.72 Å, and that each constituent

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soliton describes over 7000 supersecondary structures in the Protein Data Bank (PDB). Krokhotin, Niemi, and Peng [8] constructed a library of 200 soliton parameter sets covering over 90% of PDB loop structures at sub-ångström accuracy. More recently, the framework has been extended to thermal dynamics and structural stability modeling: Begun et al. [9] simulated the folding and unfolding of the slipknotted protein AFV3-109 using multi-soliton ansätze, and Begun et al. [10] introduced Arnold's perestroikas to characterize topological phase transitions in myoglobin. Complementing this topological perspective, Liubimov et al. [11] applied the underlying lattice Abelian Higgs model to the same myoglobin structure, demonstrating that native conformations can be stabilized by introducing heterogeneous external fields to mimic environmental interactions. These studies establish the DNLS and Abelian Higgs frameworks as a compact geometric language for characterizing known protein conformations. The Hasimoto map on which they rest is a rigorous, information-preserving change of variables, a local $U(1)$ gauge transformation of the discrete Frenet frame. These studies constitute a comprehensive treatment of the *inverse problem*: given a known structure, the soliton basis provides an efficient representation and a robust tool for analyzing structural fluctuations. The complementary *forward problem*, generating the native conformation strictly from sequence-level input, is a logically distinct question, and it is the one we take up here.

Concretely, if the DNLS effective potential $V_{\text{eff}}[n]$ could be determined from the amino-acid sequence alone, one could in principle solve the DNLS forward to obtain the native (κ, τ) profile and reconstruct the three-dimensional structure. Whether this is possible turns on whether the DNLS acts as a dynamical governing equation that drives folding or merely as a kinematic identity that records the final state.

Across theoretical biophysics, predictive success has consistently relied on capturing non-local information that the nearest-neighbor structure of the DNLS cannot represent. Energy landscape theory [12] and coarse-grained models like AWSEM [13] achieve accuracy by incorporating explicit long-range contact potentials that bias the free-energy surface. Similarly, from a geometric perspective, tube models [14,15] rely on excluded-volume interactions to select secondary structures, while direct coupling analysis (DCA) [16,17] extracts long-range contacts from evolutionary covariance. Topological features such as knots [18,19] further impose global constraints that lie beyond any local curvature description.

The same lesson emerges most clearly from recent deep-learning methods. AlphaFold 2 [20] and AlphaFold 3 [21] address the prediction problem by inferring a full rigid-body frame (rotation and translation) for every residue, effectively retaining all orientational degrees of freedom. Even single-sequence methods like ESMFold [22,23], driven by protein language models, achieve competitive accuracy by implicitly mapping the sequence into a high-dimensional embedding space that captures non-local structural and dynamic context [24]. By integrating evolutionary covariance or contextual embeddings, these methods explicitly reconstruct the global information defining the native fold. In contrast, the Hasimoto transform expresses the backbone geometry as a single complex scalar field ψ , after which the modeling step reduces the dynamics to a local effective potential. The critical unresolved issue is whether this local real-potential reduction retains sufficient information to determine the three-dimensional structure, or whether it strips away the non-local and chiral information required for folding.

In this work, we derive an exact closed-form decomposition of the DNLS effective potential $V_{\text{eff}} = V_{\text{re}} + i V_{\text{im}}$ into explicit functions of the curvature ratios $r^\pm = \kappa[n \pm 1]/\kappa[n]$ and torsion angles $\tau[n]$, and verify this decomposition to machine precision ($< 10^{-14}$) on 856 non-redundant proteins spanning all four SCOP structural classes. The decomposition reveals three independent structural barriers to forward prediction. First, the imaginary part V_{im} encodes the sign of the torsion angle through the odd symmetry of $\sin \tau$, with magnitude roughly 31% that of the real part; discarding it introduces a 2^N chiral degeneracy that grows exponentially with chain length. Second, the real part V_{re} is 95% determined by local backbone geometry rather than by the

amino-acid sequence, leaving the explicit sequence-to-potential channel too weak to carry the chemical information needed for prediction. Third, self-consistent field iterations driven by hydrophobic and elastic potentials, with and without hydrogen-bond terms, fail uniformly on all 856 proteins (mean RMSD = 13.1 Å, torsion correlation indistinguishable from zero), and the two settings produce statistically identical outcomes. Beyond identifying these static and dynamical barriers, we characterize the descriptive-predictive boundary using the language of integrable systems. We prove that within a uniform segment, the DNLS reduces to a constant-coefficient linear recurrence whose transfer matrix has unit determinant, so that the interior of the segment is rigidly fixed by its boundary data. An integrable segment can describe a backbone whose endpoints are given, but it cannot generate an interior conformation from sequence. By classifying the physical interactions governing folding by their parity under torsion inversion and their locality along the chain, we further find that the local, real-potential DNLS occupies a single (even $\times$ local) sector, whereas every driver of tertiary structure (solvent burial, electrostatics, disulfides, and hydrogen bonding) lies in the non-local row. The on-site equations underlying these reductions carry no Lax pair, which provides the structural reason that this description succeeds where *ab initio* generation cannot. These barriers are structural rather than algorithmic. They arise from the purely local, real-potential reduction of the DNLS, not from the Hasimoto map (which is lossless), and are unlikely to be circumvented by parameter tuning or improved optimization alone. At the same time, the decomposition yields a constructive result. We show that the residual of the DNLS dispersion relation serves as a purely geometric helix detector with ROC AUC = 0.72 across all SCOP classes, identifying α -helical segments as the backbone regions where the DNLS most closely approximates an integrable system. This suggests a geometric criterion for secondary-structure identification that does not require hydrogen-bond information.

The remainder of this paper is organized as follows. Section 2 introduces the discrete Frenet frame and the DNLS formulation. Section 3 presents the exact decomposition and its statistical validation. Section 4 analyzes piecewise integrability and its connection to secondary structure. Section 5 reports the self-consistent field tests. Section 6 discusses the physical origin of the barriers and the constructive applications of the framework.

2. Discrete Frenet frame and DNLS formulation

2.1. Discrete Frenet geometry of the C_α backbone

We represent the protein backbone as a polygonal chain of C_α positions $\{\mathbf{r}[n]\}_{n=1}^N$ in $\mathbb{R}^3$, where N is the number of residues. The bond vectors are defined as

$$\mathbf{t}[n] = \mathbf{r}[n+1] - \mathbf{r}[n], \quad n = 1, \dots, N-1. \quad (1)$$

For a C_α trace the bond lengths $|\mathbf{t}[n]|$ are approximately 3.8 Å [25,26], corresponding to the virtual-bond distance between consecutive α -carbons. We work with the unit tangent vectors $\hat{\mathbf{t}}[n] = \mathbf{t}[n]/|\mathbf{t}[n]|$.

The discrete Frenet frame at vertex n is constructed from three successive C_α positions. The bond angle $\kappa[n]$ is defined through

$$\cos \kappa[n] = \hat{\mathbf{t}}[n] \cdot \hat{\mathbf{t}}[n-1], \quad n = 2, \dots, N-1, \quad (2)$$

so that $\kappa[n] \in [0, \pi]$ measures the bending of the chain at vertex n . The torsion angle $\tau[n]$ requires four successive C_α positions and is defined as the dihedral angle

$$\tau[n] = \text{atan2}((\hat{\mathbf{t}}[n-1] \times \hat{\mathbf{t}}[n]) \times (\hat{\mathbf{t}}[n] \times \hat{\mathbf{t}}[n+1]) \cdot \hat{\mathbf{t}}[n], (\hat{\mathbf{t}}[n-1] \times \hat{\mathbf{t}}[n]) \cdot (\hat{\mathbf{t}}[n] \times \hat{\mathbf{t}}[n+1])), \quad (3)$$

with $\tau[n] \in (-\pi, \pi]$, defined for $n = 2, \dots, N-2$. The sign of τ encodes the local handedness of the backbone: positive values correspond to right-handed twisting and negative values to left-handed twisting.

The pair $(\kappa[n], \tau[n])$ constitutes a complete set of internal coordinates for the discrete curve. Given an initial position $\mathbf{r}[1]$, an initial tangent

$\hat{\mathbf{t}}[1]$, and an initial normal direction, the full three-dimensional backbone can be reconstructed from $\{\kappa[n], \tau[n]\}$ by sequential application of discrete Frenet rotations. This reconstruction is unique up to a global rigid-body transformation (three translations and three rotations), so that κ and τ encode all shape information of the backbone.

The discrete normal and binormal vectors are defined as

$$\hat{\mathbf{n}}[n] = \frac{\hat{\mathbf{t}}[n] - \cos \kappa[n] \hat{\mathbf{t}}[n-1]}{\sin \kappa[n]}, \quad \hat{\mathbf{b}}[n] = \hat{\mathbf{t}}[n-1] \times \hat{\mathbf{n}}[n], \quad (4)$$

forming an orthonormal triad $\{\hat{\mathbf{t}}[n-1], \hat{\mathbf{n}}[n], \hat{\mathbf{b}}[n]\}$ at each interior vertex. The torsion angle $\tau[n]$ then governs the rotation of the normal plane from vertex n to vertex $n+1$ about the tangent direction.

2.2. Hasimoto transform and complex scalar field

Following the construction introduced by Hasimoto for continuous curves, we define a complex scalar field on the backbone through

$$\psi[n] = \kappa[n] \exp\left(i \sum_{k=2}^n \tau[k]\right), \quad n = 2, \dots, N-2. \quad (5)$$

The modulus $|\psi[n]| = \kappa[n]$ records the local bending, while the phase $\arg \psi[n] = \sum_{k=2}^n \tau[k]$ accumulates the torsion along the chain. This transform maps the two real geometric fields (κ, τ) into a single complex field ψ , reducing the description of backbone geometry to a one-component problem at the cost of entangling curvature and torsion in the amplitude and phase of ψ . The gauge-theoretic program of Niemi and collaborators [3–6] works directly with the curvature and torsion fields of the discrete Frenet frame, treating κ and τ separately. We instead adopt the explicit complex-field form $\psi = \kappa e^{i \sum \tau}$ and identify it as a discrete analogue of the Hasimoto transform. This form makes manifest the amplitude-phase entanglement that underlies the present analysis.

The inverse transform recovers the geometric variables as

$$\kappa[n] = |\psi[n]|, \quad \tau[n] = \arg \psi[n] - \arg \psi[n-1]. \quad (6)$$

Given $\psi[n]$ for all interior vertices, the backbone can therefore be reconstructed by extracting (κ, τ) via Eq. (6) and applying the discrete Frenet reconstruction.

2.3. Discrete nonlinear Schrödinger equation

The discrete nonlinear Schrödinger (DNLS) equation is a standard model in nonlinear lattice dynamics [27,28]; in the version used here for the backbone field $\psi[n]$ it takes the form of a discrete eigenvalue problem

$$\beta^+[n](\psi[n+1] - \psi[n]) - \beta^-[n](\psi[n] - \psi[n-1]) = V_{\text{eff}}[n] \psi[n], \quad (7)$$

where $\beta^+[n]$ and $\beta^-[n]$ are site-dependent coupling parameters and $V_{\text{eff}}[n]$ is the effective potential. The left-hand side is a discrete second difference of ψ with nonuniform coefficients, analogous to a lattice Laplacian.

The coupling parameters $\beta^\pm[n]$ encode the local mechanical properties of the peptide chain. In the simplest formulation they depend on the virtual-bond lengths as

$$\beta^+[n] = \frac{1}{|t[n]|}, \quad \beta^-[n] = \frac{1}{|t[n-1]|}. \quad (8)$$

Because the C_α – C_α virtual-bond length varies only weakly along the chain (standard deviation $\sim 0.1 \text{ \AA}$ around the mean of $3.8 \text{ \AA}$), the coupling parameters are nearly uniform. This near-uniformity will play a central role in the analysis of Section 3, where we show that the sequence dependence of $\beta^\pm$ contributes less than 5% of the variance of V_{re} .

The effective potential is obtained by rearranging Eq. (7):

$$V_{\text{eff}}[n] = \frac{1}{\psi[n]} \left[\beta^+[n](\psi[n+1] - \psi[n]) - \beta^-[n](\psi[n] - \psi[n-1]) \right]. \quad (9)$$

This expression is well defined wherever $\psi[n] \neq 0$, i.e., wherever the bond angle $\kappa[n] \neq 0$. For physical protein backbones $\kappa[n]$ is strictly positive at all interior vertices, so V_{eff} is defined throughout the chain.

Because ψ is complex, V_{eff} is in general complex:

$$V_{\text{eff}}[n] = V_{\text{re}}[n] + i V_{\text{im}}[n]. \quad (10)$$

The real and imaginary parts carry distinct geometric content, as we demonstrate through the decomposition in Section 3.

Two features of Eq. (9) merit emphasis. First, the equation is an algebraic identity: for any discrete curve with $\kappa[n] > 0$, one can always compute ψ via Eq. (5) and then define V_{eff} via Eq. (9). No physical assumption enters this construction. The potential V_{eff} is not postulated; it is read off from the known geometry. Second, the equation becomes a dynamical equation only when V_{eff} is specified independently of the geometry, for example through a physical energy functional. In the vortex-filament context, the local induction approximation provides exactly such a specification, and the Hasimoto transform converts the geometric evolution into the integrable NLS. For proteins, the question is whether an analogous physical specification of V_{eff} exists. The analysis of the following sections indicates that the answer is negative: the mathematical form of V_{eff} on protein backbones presents structural obstacles to its determination from sequence information alone.

2.4. Connection to the continuum Hasimoto transform

In the continuum limit where the lattice spacing $\Delta s \rightarrow 0$ and the coupling becomes uniform ($\beta^\pm \rightarrow 1/\Delta s$), the discrete second difference in Eq. (7) reduces to $\partial^2 \psi / \partial s^2$, and the DNLS recovers the structure of the cubic NLS

$$i \frac{\partial \psi}{\partial t} = \frac{\partial^2 \psi}{\partial s^2} + \frac{1}{2} |\psi|^2 \psi. \quad (11)$$

The soliton solutions of Eq. (11) describe localized bending excitations that propagate without dispersion along the filament. In the protein context, the discrete soliton solutions of Eq. (7) have been shown to reproduce the (κ, τ) profiles of α -helices (dark solitons with $\kappa \approx 1.5 \text{ rad}$, $\tau \approx 1.0 \text{ rad}$) and β -strands (bright solitons with larger κ and $\tau \approx \pm \pi$). The key distinction is that for vortex filaments Eq. (11) is both kinematic (relating ψ to geometry) and dynamic (governing time evolution), whereas for proteins only the kinematic content survives. Establishing this distinction quantitatively is the purpose of the remainder of this paper.

3. Exact discrete decomposition

The effective potential $V_{\text{eff}}[n]$ defined by Eq. (7) encodes the full geometric content of the backbone in a single complex-valued sequence. In this section we derive a closed-form decomposition of V_{eff} into real and imaginary parts, each expressed entirely in terms of the local curvature ratios and torsion angles. The decomposition is an algebraic identity: it holds for any discrete space curve, independent of any physical model or approximation.

3.1. Decomposition identity

Proposition 1 (Decomposition identity). *Let $\psi[n] = \kappa[n] \exp(i \sum_{k=2}^n \tau[k])$ be the Hasimoto field constructed from the discrete Frenet curvature $\kappa[n] > 0$ and torsion $\tau[n]$, and let $V_{\text{eff}}[n]$ be defined through the discrete Schrödinger equation*

$$\beta_n^+ (\psi[n+1] - \psi[n]) - \beta_n^- (\psi[n] - \psi[n-1]) = V_{\text{eff}}[n] \psi[n], \quad (12)$$

where $\beta_n^\pm$ are the (possibly sequence-dependent) bond stiffness parameters. Then $V_{\text{eff}}[n] = V_{\text{re}}[n] + i V_{\text{im}}[n]$ with

$$V_{\text{re}}[n] = \beta_n^+ r^+[n] \cos \tau[n+1] + \beta_n^- r^-[n] \cos \tau[n] - (\beta_n^+ + \beta_n^-), \quad (13)$$

$$V_{\text{im}}[n] = \beta_n^+ r^+[n] \sin \tau[n+1] - \beta_n^- r^-[n] \sin \tau[n], \quad (14)$$

where the curvature ratios are

$$r^+[n] \equiv \frac{\kappa[n+1]}{\kappa[n]}, \quad r^-[n] \equiv \frac{\kappa[n-1]}{\kappa[n]}. \quad (15)$$

Table 1

Numerical verification of the exact decomposition on eight representative proteins (two per SCOP class). N : number of C_α atoms; ϵ : maximum absolute error between the two independent computations of V_{eff} ; $\langle |V_{\text{im}}|/|V_{\text{re}}| \rangle$: mean ratio of imaginary to real potential magnitude; ρ_{geom} : Spearman correlation between V_{re} computed with physical $\beta(s)$ and with uniform $\beta = 1$; δ_H , δ_E , δ_C : dispersion-relation RMSE (in degrees) for helix, strand, and coil residues respectively; RMSD: backbone RMSD (Å) of the structure reconstructed from V_{re} only (setting $V_{\text{im}} = 0$). A dash indicates that the corresponding secondary-structure type is absent.

PDB	Class	N	ϵ	$\langle V_{\text{im}} / V_{\text{re}} \rangle$	ρ_{geom}	δ_H	δ_E	δ_C	RMSD
1PA7	all- α	130	4.9×10^{-15}	0.324	0.966	20.4	—	39.4	51.2
2RH3	all- α	130	3.1×10^{-15}	0.332	0.963	27.1	36.6	35.6	32.5
5SV5	all- β	134	1.4×10^{-14}	0.290	0.913	—	41.7	40.0	28.9
1AYO	all- β	130	1.8×10^{-14}	0.247	0.971	20.1	37.1	40.3	56.6
2B1L	α/β	129	5.1×10^{-15}	0.300	0.945	17.8	38.2	42.5	55.9
1JBE	α/β	132	1.8×10^{-15}	0.305	0.956	21.3	45.8	28.8	36.6
3CIP	$\alpha + \beta$	130	9.3×10^{-15}	0.273	0.949	22.1	31.1	36.3	55.4
2R4I	$\alpha + \beta$	130	3.6×10^{-15}	0.264	0.969	24.6	39.4	32.7	48.2

Proof. We substitute the Hasimoto ansatz into Eq. (12) and divide both sides by $\psi[n] \neq 0$, which gives

$$\frac{V_{\text{eff}}[n] \psi[n]}{\psi[n]} = \beta_n^+ \left(\frac{\psi[n+1]}{\psi[n]} - 1 \right) - \beta_n^- \left(1 - \frac{\psi[n-1]}{\psi[n]} \right). \quad (16)$$

The ratio of adjacent Hasimoto fields is

$$\frac{\psi[n+1]}{\psi[n]} = \frac{\kappa[n+1]}{\kappa[n]} \exp(i \tau[n+1]) = r^+[n] e^{i\tau[n+1]}, \quad (17)$$

and similarly

$$\frac{\psi[n-1]}{\psi[n]} = \frac{\kappa[n-1]}{\kappa[n]} \exp(-i \tau[n]) = r^-[n] e^{-i\tau[n]}. \quad (18)$$

Collecting terms,

$$V_{\text{eff}}[n] = \beta_n^+ (r^+[n] e^{i\tau[n+1]} - 1) - \beta_n^- (1 - r^-[n] e^{-i\tau[n]}). \quad (19)$$

Separating real and imaginary parts via Euler's formula yields Eqs. (13)-(14) directly. $\square$

The derivation is algebraically direct; the value of stating it explicitly lies in disentangling the roles of $r^\pm$ and τ that are obscured in the complex field ψ , and in enabling the systematic tests of Sections 3–5.

Two further features of this decomposition merit emphasis. First, the result is exact: no continuum limit, Taylor expansion, or slowly varying envelope approximation has been invoked. Second, the decomposition holds for arbitrary bond parameters $\beta_n^\pm$ and for any discrete curve with $\kappa[n] > 0$; it is a kinematic identity rather than a dynamical equation.

3.2. Numerical verification

We verify the decomposition on eight representative proteins spanning the four SCOP [29,30] structural classes (Table 1). For each protein, we compute $V_{\text{eff}}[n]$ in two independent ways: (i) directly from the definition Eq. (12) using the Hasimoto field $\psi[n]$, and (ii) from the analytic expressions Eqs. (13)-(14) using $\kappa[n]$ and $\tau[n]$. The maximum absolute difference over all residues is reported in the column labeled ϵ in Table 1.

In all eight cases the error ϵ is of order 10^{-15} to 10^{-14} , consistent with IEEE 754 double-precision arithmetic. This confirms that Eqs. (13)-(14) reproduce V_{eff} to machine precision and contain no hidden approximation. The remaining columns of Table 1 preview the structural barriers analyzed in the following subsections and in Sections 4–5; we defer their full statistical treatment to the analysis of the 856-protein dataset.

3.3. Barrier I: imaginary potential and torsion-sign ambiguity

The imaginary part V_{im} (Eq. (14)) depends on τ exclusively through the sine function. Because $\sin(\tau)$ is odd, V_{im} changes sign under $\tau \rightarrow -\tau$

while V_{re} (Eq. (13)), which depends on $\cos(\tau)$, remains invariant. We now formalize the resulting degeneracy as a symmetry statement. This result holds *within the class of purely local, real-potential models*, where the modeling choice is to retain only $V_{\text{re}} = \text{Re } V_{\text{eff}}$. The decomposition (Eqs. (13)-(14)) is itself an exact kinematic identity that loses no information; the degeneracy below is a consequence of *discarding* V_{im} , and is therefore a property of the model class, not of the Hasimoto map.

Definition 1 (Site-wise torsion-parity group). Let N be the chain length, i.e. the number of torsion angles $\tau[1], \dots, \tau[N]$. Let $G = (\mathbb{Z}_2)^N$, whose elements are subsets $S \subseteq \{1, \dots, N\}$ with group product the symmetric difference, generated by the elementary flips $g_k = \{k\}$. G acts on the torsion configuration by site-wise inversion, $(g_S \cdot \tau)[k] = -\tau[k]$ if $k \in S$ and $\tau[k]$ otherwise.

Lemma 1 (Site-wise torsion-parity symmetry and chiral degeneracy). Fix the curvatures $\kappa[n] > 0$ and the real bond parameters $\beta_n^\pm$. For every $g_S \in G = (\mathbb{Z}_2)^N$ the real potential is invariant,

$$V_{\text{re}}[n](g_S \cdot \tau) = V_{\text{re}}[n](\tau) \quad (n = 1, \dots, N), \quad (20)$$

i.e. each $V_{\text{re}}[n]$ carries the trivial representation of G , whereas every term of V_{im} containing $\tau[k]$ transforms by the sign character $\chi_k(g_S) = (-1)^{|k \in S|}$ and thus changes sign under the generator g_k . Consequently, for an open (terminated) chain the configurations sharing a given profile $\{V_{\text{re}}[n]\}_{n=1}^N$ and related by site-wise torsion-sign flips form a single G -orbit, of size

$$\#(G \cdot \tau) = 2^{N'}, \quad N' = \#\{k : \tau[k] \notin \{0, \pi\}\}. \quad (21)$$

For a chirally generic backbone ($\tau[k] \notin \{0, \pi\}$ for all k) this is exactly 2^N .

Proof. Since $G = \langle g_1, \dots, g_N \rangle$ with commuting generators, it suffices to treat a single generator g_k . By Eq. (13) the angle $\tau[k]$ appears in V_{re} only in $V_{\text{re}}[k]$ (through the term $\beta_{k-1}^- r^-[k] \cos \tau[k]$) and in $V_{\text{re}}[k-1]$ (through the term $\beta_{k-1}^+ r^+[k-1] \cos \tau[(k-1)+1] = \beta_{k-1}^+ r^+[k-1] \cos \tau[k]$); no other entry depends on $\tau[k]$, and κ , $r^\pm$, $\beta^\pm$ are independent of the torsion sign. As g_k sends $\tau[k] \mapsto -\tau[k]$ and $\cos$ is even, both entries are unchanged, proving invariance. In Eq. (14) the same two slots carry $\tau[k]$ through the odd function $\sin$, so those terms change sign; this is the sign character χ_k . The generators act on disjoint coordinates and commute, so the statements extend to an arbitrary g_S by composition. The orbit count (21) then follows from the orbit-stabilizer theorem: a flip at a site with $\tau[k] \in \{0, \pi\}$ is trivial (there $\sin \tau[k] = 0$ and $\tau[k]$ lies on a symmetry axis of $\cos$), so $|\text{Stab}_G(\tau)| = 2^{N-N'}$ and $\#(G \cdot \tau) = |G|/|\text{Stab}_G(\tau)| = 2^{N'}$. $\square$

The sign of $\tau[k]$ is the local chirality. Lemma 1 shows that retaining only the real potential projects every observable onto the G -invariant (trivial-representation) subspace; the chiral information resides entirely in the sign sectors carried by V_{im} and is annihilated by this projection. Equivalently, discarding V_{im} is exactly the projection onto the G -invariants. A right-handed α -helix and its per-residue left-handed enantiomer therefore produce identical V_{re} profiles, and reconstruction from V_{re} alone confronts the 2^N -fold ambiguity of Eq. (21) for a generic backbone. This is the rigorous content of the statement that a purely real potential cannot break the chiral symmetry; restriction to V_{re} is identical to projection onto the G -invariants, which are insensitive to the sign sectors.

This site-wise parity $\tau[k] \rightarrow -\tau[k]$ is a distinct object from the discrete $\mathbb{Z}_2$ gauge transformation $\kappa[k] \rightarrow \kappa[k] \cos \Delta_k$, $\tau[k] \rightarrow \tau[k] + \Delta_{k-1} - \Delta_k$ (with $\Delta_k \in \{0, \pi\}$) introduced by Chernodub et al. [4] to maintain the convention $\kappa \geq 0$ across conjugacy points. The latter leaves the Cartesian coordinates intact and is therefore a genuine gauge redundancy, whereas the parity considered here changes the backbone chirality and hence the three-dimensional structure, a change rendered invisible only by discarding V_{im} .

The degeneracy can be lifted only by adding to the energy a term that is *odd* under each g_k , i.e. that has a nonzero component in the sign sectors. The minimal local choice is a sum of single-site odd functions, $H_{\text{CS}}[\tau] = \lambda \sum_{k=1}^N f(\tau[k])$ with $f(-\theta) = -f(\theta)$, such as the discrete total-torsion (Chern–Simons-type) coupling $\lambda \sum_k \tau[k]$. Under g_S one finds

$H_{CS}[g_S \cdot \tau] = H_{CS}[\tau] - 2\lambda \sum_{k \in S} f(\tau[k])$, so for $\lambda \neq 0$ no nontrivial g_S remains a symmetry: the group is explicitly broken and minimization selects a unique handedness at each site, with $\text{sign}(\lambda)$ fixing the global chirality. This is consistent with the observation of Danielsson et al. [3] that the torsion sign can be fixed by a Chern–Simons term; we emphasize, however, that such a term *injects* the missing chirality information rather than conferring predictive power, and the degeneracy is a property of the local, real-potential model class, not of the Hasimoto map itself.

Remark (discrete probability current). The role of V_{im} can be stated as a conservation law. Multiplying Eq. (12) by $\bar{\psi}[n]$ and taking the imaginary part gives, for arbitrary site-dependent real $\beta_n^\pm$ and complex V_{eff} , the exact discrete continuity identity

$$J[n] - J[n-1] = V_{\text{im}}[n] |\psi[n]|^2, \quad (22)$$

$$J[n] = \beta_n^+ \text{Im}(\bar{\psi}[n]\psi[n+1]) = \beta_n^+ \kappa[n]\kappa[n+1] \sin \tau[n+1],$$

where $J[n]$ is the discrete probability current (the $(\psi, \bar{\psi})$ Casoratian of the recurrence). Thus V_{im} is the *unique source* of J : $V_{\text{im}} = 0$ is precisely the condition under which J would be conserved along the chain. Real backbones have generically nonzero V_{im} (median $\langle |V_{\text{im}}|/|V_{\text{re}}| \rangle \approx 0.30$ over the 856 chains; nonzero at 99.8% of sites), so the geometric flux J is genuinely *not* conserved, changing sign along every chain, and we make no claim that real proteins obey $\kappa \kappa' \sin \tau = \text{const.}$ The consequence is that restricting to a real effective potential is not a constraint imposed on the backbone but a *model class* that is structurally blind to this flux: it discards the single gauge-covariant, parity-odd field V_{im} that sources the only nontrivial conserved bilinear current of the discrete map. The torsion-sign degeneracy of Lemma 1 is the kinematic statement of this fact, and Eq. (22) is its dynamical counterpart. The discarded imaginary part is at once the carrier of chirality and the source of the probability current.

We quantify the information content of V_{im} across the full dataset of 856 non-redundant proteins. Fig. 1(a) shows the distribution of the ratio $\langle |V_{\text{im}}|/|V_{\text{re}}| \rangle$ per protein. The mean ratio is 0.31, indicating that the imaginary component has a magnitude roughly one-third that of the real part. The distribution is largely independent of SCOP class, with all- α proteins showing a slightly broader tail toward higher values.

To demonstrate the practical consequence of discarding V_{im} , we reconstruct the backbone of each protein using V_{re} only (setting $V_{\text{im}} = 0$ and choosing $\text{sign}(\tau)$ uniformly positive). Fig. 1(b) plots the resulting C α RMSD against chain length. The RMSD grows approximately linearly, reaching 40–120 Å for chains of 200–300 residues. This linear growth reflects the cumulative nature of torsion-sign errors: each incorrectly assigned sign produces a local angular deviation that propagates along the chain. In information-theoretic terms, each residue contributes one bit of unresolved chiral information, for a total of N bits per chain. The V_{re} -only reconstruction therefore yields structures that deviate substantially from the native fold for all but the shortest peptides.

3.4. Barrier II: geometric dominance of V_{re}

The decomposition (Eq. (13)) shows that V_{re} depends on the bond parameters $\beta_n^\pm$ only as multiplicative prefactors of the geometric terms $r^\pm \cos \tau$. A natural question is whether the sequence dependence encoded in $\beta(s)$ contributes significantly to the spatial profile of V_{re} , or whether the geometric factors dominate.

We test this by computing, for each of the 856 proteins, two versions of V_{re} : one with the physical, amino-acid-dependent bond parameters $\beta_n^\pm(s)$, and one with uniform $\beta^+ = \beta^- = 1$. The Spearman rank correlation ρ_{geom} between the two profiles measures the fraction of the V_{re} pattern that is determined by geometry alone.

Corollary 1 (Geometric dominance). *Across 856 non-redundant proteins, the Spearman correlation between $V_{\text{re}}(\beta(s))$ and $V_{\text{re}}(\beta = 1)$ has mean $\bar{\rho}_{\text{geom}} = 0.951$ and minimum 0.88 (Fig. 2). The residual variance attributable*

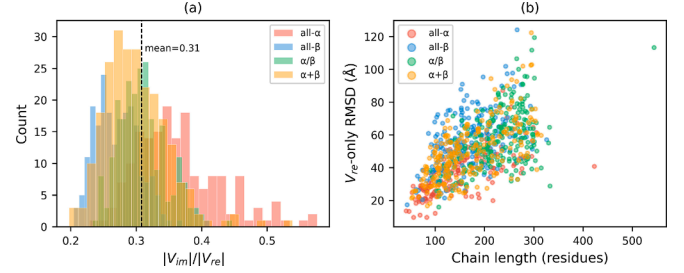


Fig. 1. Information cost of discarding the imaginary potential. (a) Distribution of the ratio $\langle |V_{\text{im}}|/|V_{\text{re}}| \rangle$ across 856 non-redundant proteins, colored by SCOP class. The mean ratio is 0.31, indicating that the imaginary component, which encodes the sign of the torsion angle through the odd symmetry of $\sin(\tau)$, has a magnitude roughly one-third that of the real part. The distribution is largely class-independent, with all- α proteins showing a slightly broader tail toward higher values. (b) Backbone RMSD of structures reconstructed using V_{re} only (setting $V_{\text{im}} = 0$) versus chain length. RMSD grows roughly linearly with chain length, reaching 40–120 Å for chains of 200–300 residues. This reflects the cumulative effect of torsion-sign errors: each residue contributes 1 bit of unresolved chiral ambiguity, and the resulting 2^N degeneracy renders V_{re} -only reconstruction structurally unreliable beyond short peptides.

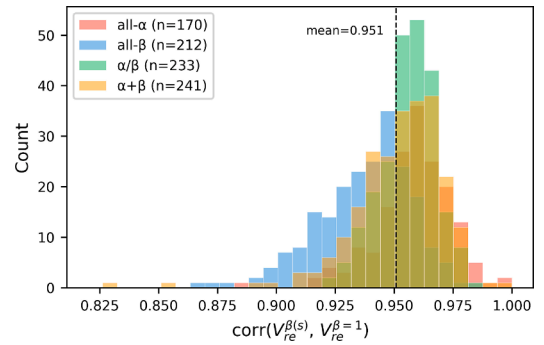


Fig. 2. Geometric dominance of the real effective potential. Distribution of Spearman rank correlation between V_{re} computed with the physical, sequence-dependent bond parameters $\beta(s)$ and V_{re} computed with uniform $\beta = 1$, evaluated over 856 non-redundant proteins. Colors denote SCOP structural classes: all- α (170), all- β (212), α/β (233), and $\alpha + \beta$ (241). The distribution is sharply peaked near unity (mean = 0.951), indicating that the explicit sequence dependence carried by $\beta(s)$ accounts for less than 5% of the variance of V_{re} on average. The dominant contribution comes from the geometric terms $r^\pm \cos \tau$, which depend on the backbone structure (κ, τ) rather than directly on amino-acid identity. All four SCOP classes overlap, confirming that this pattern is universal across protein folds.

to sequence-dependent $\beta(s)$ is less than $1 - \bar{\rho}^2 \approx 0.05$, i.e., below 5% on average.

This finding indicates that sequence information is decoupled from the effective potential. Although the bond parameters $\beta_n^\pm$ depend on amino-acid identity, their contribution to the variance of V_{re} is negligible. The potential is dominated by the geometric terms $r^\pm \cos \tau$, the fluctuations of which exceed those induced by sequence-dependent stiffness by an order of magnitude (Fig. 4). All four SCOP classes overlap in the ρ_{geom} distribution (Fig. 2), confirming that this pattern is universal across fold types. One caveat applies. The geometric terms $r^\pm$ and τ are themselves determined by the amino-acid sequence through the folding process. The Spearman test does not show that V_{re} is independent of sequence in an absolute sense; rather, it establishes that the only *explicit* pathway from sequence to V_{re} within the decomposition (Eqs. (13)–(14)), namely the bond stiffnesses $\beta_n^\pm$, is too narrow to carry significant information. The sequence dependence of V_{re} is almost entirely *implicit*, mediated by the three-dimensional structure (κ, τ) that the folding process produces.

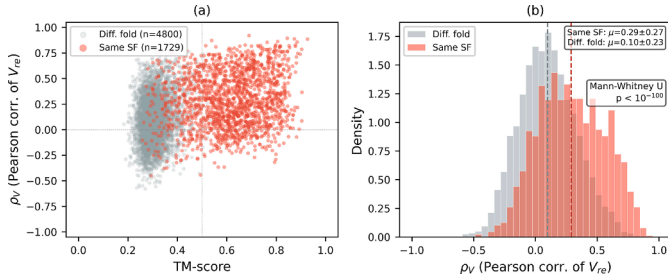


Fig. 3. V_{re} tracks fold rather than sequence. (a) Pearson correlation ρ_V of structurally aligned V_{re} profiles versus TM-score for 1729 same-superfamily pairs (red) and 4800 different-fold pairs (gray). Among the same-superfamily pairs, 79% have TM-score > 0.5 and cluster in the upper-right quadrant; the remaining 21% are distant homologs with greater structural divergence, yet ρ_V still correlates positively with TM-score within this subgroup. (b) Distribution of ρ_V for the two groups. Same-superfamily: $\mu = 0.29 \pm 0.27$; different-fold: $\mu = 0.10 \pm 0.23$ (Mann-Whitney U , $p < 10^{-134}$). The mean sequence identity within the same-superfamily group is 13.9%, confirming that the elevated correlation is driven by structural similarity, not sequence similarity.

To test this interpretation directly, we compared V_{re} profiles between protein pairs that share the same SCOP superfamily but have low sequence identity. From the 856 non-redundant proteins (culled by the PISCES [31] server with resolution $\leq 2.0 \text{ \AA}$, R -factor ≤ 0.2 , chain length 40–300 residues, sequence identity $\leq 25\%$, X-ray entries only, excluding chains with breaks or disorder), we identified 1729 pairs belonging to the same SCOP superfamily. As a control, we constructed 4800 pairs drawn from different SCOP folds with chain-length differences ≤ 20 residues. For each pair, TM-align was used to obtain a residue-level structural alignment; the Pearson correlation ρ_V was then computed between the V_{re} values at structurally aligned positions.

Fig. 3 summarizes the results. Same-superfamily pairs exhibit a mean $\rho_V = 0.290 \pm 0.266$, significantly higher than the different-fold background of $\rho_V = 0.099 \pm 0.230$ (Mann-Whitney U test, $p < 10^{-134}$). The scatter plot Fig. 3(a) shows that ρ_V increases with TM-score: among the 1729 same-superfamily pairs, 79% have TM-score > 0.5 and cluster in the upper-right quadrant, while different-fold pairs remain near zero. The remaining 21% of same-superfamily pairs fall below TM-score $= 0.5$; these are distant homologs for which the superfamily-level classification permits substantial structural divergence. Even within this subgroup, ρ_V correlates positively with TM-score, indicating that the relationship between structural similarity and V_{re} similarity is continuous rather than threshold-dependent. The mean sequence identity within the same-superfamily group is only 13.9% (range 1.4–27.9%), confirming that the elevated ρ_V is driven by structural similarity rather than sequence similarity. This provides direct evidence that V_{re} tracks the three-dimensional fold: proteins with unrelated sequences but similar structures produce similar V_{re} profiles, whereas proteins with comparable chain lengths but different folds do not.

The exact decomposition (Eq. (13)) establishes that V_{re} is driven by two distinct inputs: the geometric terms $r^\pm \cos \tau$, which depend on the three-dimensional structure, and the bond stiffnesses $\beta_n^\pm$, which depend directly on the amino-acid sequence. However, the Spearman test demonstrates that the sequence-dependent terms contribute less than 5% of the variance, and the superfamily analysis (Fig. 3) confirms that V_{re} covaries with the fold rather than the sequence. Thermodynamically, the native state is determined by the chemical identity of the amino-acid sequence. In the DNLS formulation, this causal relationship is effectively inverted: the spatial geometry dominates the effective potential, suppressing the explicit contribution of amino-acid chemistry. This structural circularity limits forward prediction: V_{re} is overwhelmingly determined by the target backbone geometry (κ, τ) one seeks to predict, while the direct pathway from sequence to V_{re} through $\beta_n^\pm$ is mathematically too weak to carry the necessary folding information.

The circularity exposed by Barrier II can be contrasted with physical models that have achieved predictive success precisely by incorporating non-local, sequence-dependent information through channels absent from the Hasimoto decomposition. Coarse-grained force fields such as AWSEM [13] supplement local backbone terms with explicit contact potentials that couple residue pairs separated by tens to hundreds of positions along the sequence, and with associative-memory terms that bias the energy landscape toward known structural motifs. The tube model of Banavar and Maritan [14,15] derives secondary-structure selection from a three-body excluded-volume interaction that is non-local by construction. Statistical approaches based on direct coupling analysis [16,17] extract residue–residue contact maps from evolutionary covariance using the maximum-entropy principle, providing non-local structural constraints derived entirely from sequence data. In each case, the predictive power arises from terms that couple residues at positions $|m - n| \gg 1$, which constitutes precisely the type of information that the DNLS effective potential, constructed from nearest-neighbor ratios $r^\pm[n]$ and local torsion angles $\tau[n]$, cannot encode. The geometric dominance of V_{re} is therefore not merely a quantitative observation but reflects a structural mismatch between the local, kinematic content of the nearest-neighbor DNLS reduction and the non-local, thermodynamic content required for folding.

3.5. Classification of folding forces by parity and locality

Barriers I and II locate the limitation of the real-potential DNLS in two algebraic properties of the decomposition: the chiral information resides in the parity-odd part V_{im} , and the only explicit sequence channel in the parity-even part V_{re} is the near-uniform nearest-neighbor stiffness $\beta_n^\pm$. These two axes, parity under $\tau \rightarrow -\tau$ and locality along the chain, organize the physical interactions that drive folding into the four sectors of Table 2. The placement of each interaction follows from an elementary symmetry argument: a term contributes to V_{im} (is parity-odd) only if it changes sign under $\tau \rightarrow -\tau$, which requires explicit chirality; and it is non-local only if it couples residues at sequence separation $|m - n| > 1$. Site-wise inversion $\tau \rightarrow -\tau$ is a reflection isometry that preserves all interatomic distances $|\mathbf{r}_i - \mathbf{r}_j|$, so every distance-dependent interaction (Coulomb, dispersion, disulfide, π -stacking, solvent burial) is parity-even and cannot, by itself, break the chiral degeneracy of Barrier I; only the stereochemistry of the L-amino-acid side chain (the C_β branch and the C_α improper dihedral) supplies a parity-odd, $h \sin \tau$ -type term.

Three consequences follow. First, the current model populates only the evenxlocal cell; the chiral sector is a *local* addition (restoring V_{im} , or equivalently a Chern–Simons-type $\sum_k \tau[k]$ term as in Barrier I) and is therefore within reach of the framework. Second, and most important, every interaction that selects *tertiary* structure (hydrophobic burial, electrostatics, disulfide cross-links, dispersion, and the $i \rightarrow i + 4$ hydrogen bond that stabilizes the helix) occupies the non-local row, which a potential built from nearest-neighbor ratios $r^\pm[n]$ cannot encode without promotion to a range- L operator. This is the structural content of the requirement that a predictive model incorporate interactions with the external environment: in the present classification those interactions are precisely the non-local, even-parity sector (solvent and electrostatics), absent from the local DNLS by construction rather than by parameter choice. Third, among these non-local forces the bare Coulomb interaction is distinguished: its $1/r$ lattice dispersion $\sum_{m \geq 1} \cos(qm)/m = -\ln[2 \sin(q/2)]$ is the unique *non-analytic* case among the forces examined, since a $1/r^p$ tail yields a non-analytic long-wavelength dispersion if and only if $p \leq 1$, and only Coulomb saturates this bound. Physiologically, however, Debye screening cuts the divergence at $\lambda_D \approx 10 \text{ \AA}$, comparable to the chiral correlation length $\xi \approx 4$ residues, so screened electrostatics sits *at the boundary* between genuinely long-range and effectively short-range rather than dominating the global modes.

A uniform integrable segment survives only while the couplings are translation-invariant. Promoting any non-local sector to a sequence-

Table 2

Classification of the principal interactions governing protein structure by parity under site-wise torsion inversion $\tau \rightarrow -\tau$ (columns) and by locality along the backbone (rows). The current local, real-potential DNLS occupies only the even $\times$ local cell. Chirality requires the parity-odd column (V_{im} , a local addition); every driver of tertiary structure lies in the non-local row, which a nearest-neighbor scalar potential cannot represent.

	Even (V_{re})	Odd (V_{im} , chiral)
Local	backbone bending κ ; van der Waals repulsion; excluded volume; ω planarity (<i>current DNLS</i>)	C_β stereochemistry; C_α improper ($h \sin \tau$)
Non-local	solvent burial; Coulomb / salt bridges; disulfides; London dispersion; π /cation/metal; hydrogen bonds	chiral non-local packing

dependent, range- L operator, as a predictive model must, renders the effective couplings site-dependent and destroys the uniform, nearest-neighbor structure on which the soliton reductions of the DNLS rest (Section 4): the integrable reduction and the non-local content required for folding are mutually exclusive within this scalar framework. The local DNLS is thus a faithful kinematic descriptor of backbone geometry, not an *ab initio* folding engine; the forces that fold the chain live in sectors it does not reach, and reaching them forfeits the integrable structure that makes it tractable.

The exact decomposition has, in summary, revealed two independent, static barriers to forward structure prediction: the 2^N -fold torsion-sign degeneracy encoded in V_{im} (Barrier I) and the geometric dominance of V_{re} that leaves less than 5% of its variance attributable to amino-acid identity (Barrier II). Both barriers follow from the algebraic structure of the real-potential decomposition (Eqs. (13)-(14)), specifically from discarding V_{im} and from the narrowness of the $\beta_n^\pm$ channel, rather than from the Hasimoto transform itself, which preserves the full (κ, τ) information. Whether the DNLS can nevertheless function as a dynamical equation that drives folding through a physically motivated effective potential is tested in Section 5.

4. Piecewise integrability and secondary structure

The exact decomposition of Section 3 holds for arbitrary discrete curves. In this section we examine the special case of backbone segments where the curvature varies slowly, so that $r^\pm[n] \approx 1$. Under this condition the full decomposition simplifies to a scalar dispersion relation that connects V_{re} directly to the torsion angle. We show that this dispersion relation is satisfied almost exclusively within α -helical segments, providing a purely geometric criterion for secondary-structure identification.

4.1. Uniform-segment dispersion relation

When the curvature ratios satisfy $r^+[n] \approx r^-[n] \approx 1$ and the bond parameters are approximately uniform ($\beta_n^+ \approx \beta_n^- \approx \beta$), the real part of the effective potential (Eq. (13)) reduces to

$$V_{\text{re}}[n] \approx \beta [\cos \tau[n+1] + \cos \tau[n]] - 2\beta. \quad (23)$$

If the torsion angle is also locally constant ($\tau[n+1] \approx \tau[n] \approx \tau$), this further simplifies to the dispersion relation

$$\cos \tau = 1 + \frac{V_{\text{re}}}{2\beta}. \quad (24)$$

Eq. (24) is the discrete analogue of the continuum NLS dispersion relation $\omega = k^2$, expressed in terms of the backbone torsion angle. It provides a one-to-one mapping between V_{re} and $|\tau|$ within any segment where κ and τ are approximately uniform. Note that the cosine function renders Eq. (24) insensitive to the sign of τ , consistent with the torsion-sign degeneracy identified in Section 3.3.

Transfer-matrix interpretation. The uniform-segment reduction has a dynamical-systems reading that makes the integrability structure explicit. Writing the constant-coefficient recurrence $\psi[n+1] = (2 + V_{\text{re}}/\beta)\psi[n] - \psi[n-1]$ for the state $\Psi_n = (\psi[n], \psi[n-1])^T$,

$$\Psi_{n+1} = T \Psi_n, \quad T = \begin{pmatrix} 2 + \frac{V_{\text{re}}}{\beta} & -1 \\ 1 & 0 \end{pmatrix} \in SL(2, \mathbb{R}). \quad (25)$$

Since $\det T = 1$, the map is area-preserving; its eigenvalues form the reciprocal pair $e^{\pm i q}$ with the Bloch dispersion $\cos q = \frac{1}{2} \text{Tr } T$, i.e. $\cos q = 1 + V_{\text{re}}/2\beta$, identical to Eq. (24). For the helical value $V_{\text{re}} = 2\beta(\cos \tau - 1)$, which is Eq. (24) rearranged, this reduces to $\cos q = \cos \tau$, so the Bloch quasi-momentum coincides with the torsion angle in magnitude, $|q| = |\tau|$, within the band $-4\beta \leq V_{\text{re}} \leq 0$. A uniform α -helix is then an *elliptic* fixed point of T ($|\text{Tr } T| < 2$, q real), about which the segment is a marginally stable Bloch mode; the band edges $q \in \{0, \pi\}$ ($V_{\text{re}} = 0$ or -4β) are the parabolic transitions beyond which the fixed point becomes hyperbolic.

The identification is necessarily unsigned. The transfer matrix (25) is built from the real potential V_{re} alone, which enters only through $\cos \tau$ and is invariant under the site-wise parity $\tau \rightarrow -\tau$ of Lemma 1; its eigenpair $\{e^{+iq}, e^{-iq}\}$ likewise contains both signs, so neither the dispersion relation nor T selects sign q . The signed phase $\psi[n+1]/\psi[n] = e^{i\tau}$ of the full Hasimoto field is recovered only through the imaginary potential V_{im} , and the residual sign of q is precisely the chiral degree of freedom of Barrier I. The transfer-matrix analysis thus independently recovers and reinforces Lemma 1: the local linear integrability of a helical segment fixes the quasi-momentum only up to $|q| = |\tau|$, so the chiral degeneracy is woven into the integrable structure itself.

The conditions under which Eq. (24) holds are precisely the conditions that define an integrable segment of the DNLS: uniform curvature ($r^\pm \approx 1$), uniform torsion, and uniform coupling. Deviations from these conditions break integrability and cause the full decomposition (Eqs. (13)-(14)) to differ from the dispersion relation. The magnitude of this deviation therefore serves as a local measure of integrability.

4.2. Integrability error as a structural probe

We define the integrability error at each residue as

$$E[n] = \left| \cos \tau[n] - \left(1 + \frac{V_{\text{re}}[n]}{2\beta} \right) \right|, \quad (26)$$

where $\beta = \langle \beta_n^\pm \rangle$ is the chain-averaged coupling parameter. By construction, $E[n] = 0$ when the backbone at residue n satisfies the uniform-segment dispersion relation exactly, and $E[n] > 0$ when the local curvature or torsion varies too rapidly for the integrable approximation to hold.

Fig. 4 shows $V_{\text{eff}}[n]$ along the backbone for eight representative proteins (two per SCOP class); we treat the effective potential as a site-dependent profile analogous to spectral signal representations of protein sequences [32]. Within helical segments (pink background shading), V_{re} forms near-constant negative plateaus (typically -0.5 to $-1.5 \text{ \AA}^{-1}$), consistent with Eq. (24) evaluated at the canonical helix torsion $\tau_{\text{helix}} \approx 1.0 \text{ rad}$. The amplitude of V_{im} is reduced relative to loop and strand regions but remains finite. In β -strand and coil regions, both V_{re} and V_{im} fluctuate strongly (amplitudes reaching $3\text{--}6 \text{ \AA}^{-1}$), reflecting rapid residue-to-residue variation of $r^\pm$ and τ . Sharp negative spikes in V_{re} mark transitions between secondary-structure elements. These patterns are local rather than class-dependent: helical segments in the nominally all- β protein 1AYO display the same plateau behavior as those in the all- α proteins.

4.3. Statistical validation on 856 proteins

We evaluate the dispersion-relation RMSE separately for helix (H), strand (E), and coil (C) residues across the full dataset of 856 non-

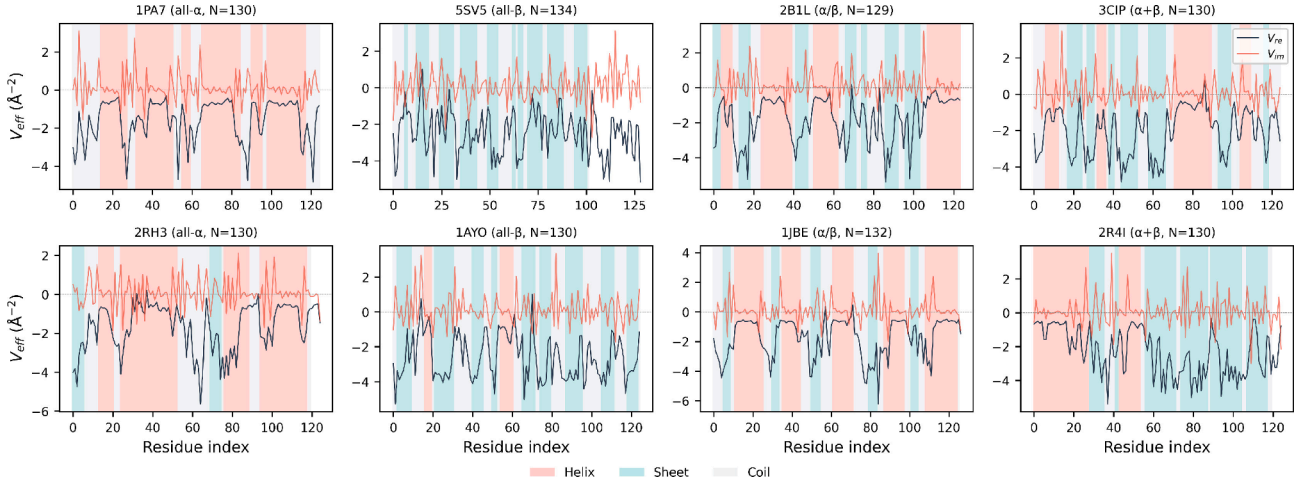


Fig. 4. Effective potential $V_{\text{eff}}[n]$ along the C_α backbone for eight representative proteins (two per SCOP class; columns from left to right: all- α , all- β , α/β , $\alpha + \beta$). Black: V_{re} ; red: V_{im} . Background shading marks DSSP secondary-structure assignment (pink: helix; cyan: strand; gray: coil). Within helical segments V_{re} forms near-constant negative plateaus consistent with the dispersion relation Eq. (24), while strand and coil regions exhibit large-amplitude fluctuations in both components. Sharp negative spikes in V_{re} mark transitions between secondary-structure elements. These patterns are local rather than class-dependent: helical segments in the all- β protein 1AYO display the same plateau behavior as those in the all- α proteins.

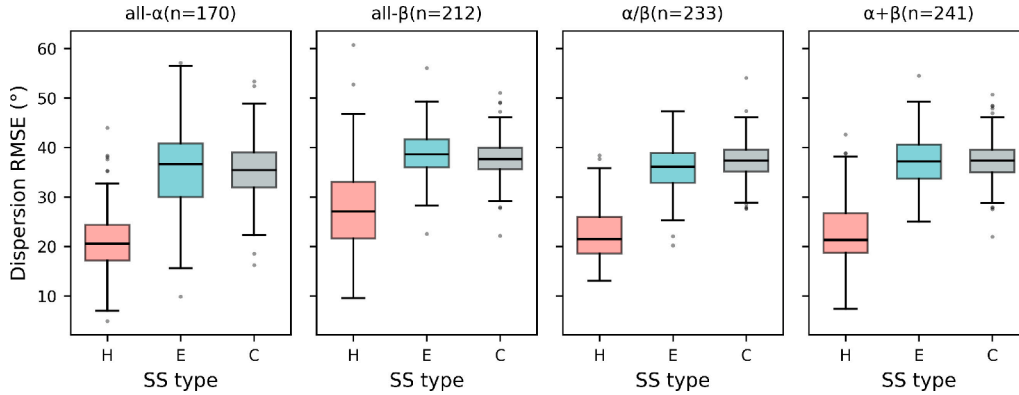


Fig. 5. Dispersion-relation RMSE by secondary-structure type, faceted by SCOP class (856 non-redundant proteins). For each protein, residues are grouped by DSSP assignment into helix (H), strand (E), and coil (C), and the RMSE of the uniform-segment approximation $\cos \tau = 1 + V_{\text{re}}/2\beta$ is computed per group. Helical segments exhibit systematically lower RMSE (median $\sim 21^\circ$) than strand ($\sim 37^\circ$) or coil ($\sim 36^\circ$) segments across all four SCOP classes. This separation is class-independent: even in all- β proteins where helices are scarce, the few helical residues satisfy the dispersion relation with comparable accuracy.

redundant proteins. For each protein, residues are grouped by their DSSP secondary-structure assignment [33], and the RMSE of Eq. (24) is computed per group:

$$\delta_X = \sqrt{\frac{1}{N_X} \sum_{n \in X} \left[\cos \tau[n] - \left(1 + \frac{V_{\text{re}}[n]}{2\beta} \right) \right]^2}, \quad (27)$$

where $X \in \{H, E, C\}$ and N_X is the number of residues in class X .

Fig. 5 shows the distribution of δ_X as box plots, faceted by SCOP class. Across all four classes, helical segments exhibit systematically lower RMSE (median $\sim 21^\circ$) than strand ($\sim 37^\circ$) or coil ($\sim 36^\circ$) segments. This separation is class-independent: even in all- β proteins where helices are scarce, the few helical residues still satisfy the dispersion relation with comparable accuracy. The result reflects the local uniformity of (κ, τ) within helices ($\tau^\pm \approx 1$), which is the condition under which the exact decomposition reduces to Eq. (24).

The data in Table 1 illustrate this pattern at the level of individual proteins. The pure all- α protein 1PA7 has $\delta_H = 20.4^\circ$ and no strand residues, while the pure all- β protein 5SV5 has $\delta_E = 41.7^\circ$ and no helix residues. In mixed-class proteins, the helix and strand RMSE values coexist within the same chain: 1AYO (all- β by SCOP classification) contains a small number of helical residues with $\delta_H = 20.1^\circ$, comparable to the values observed in all- α proteins. This confirms that the applicability of

the dispersion relation is determined by the local curvature uniformity at each residue, not by the global fold classification.

4.4. Helix detection by integrability error

The systematic separation of δ_H from δ_E and δ_C suggests that the integrability error $E[n]$ (Eq. (26)) can serve as a binary classifier for helical residues. To quantify this, we construct a receiver operating characteristic (ROC) curve by varying a threshold E_{th} and classifying residue n as helical if $E[n] < E_{\text{th}}$ and as non-helical otherwise. The ground truth is provided by DSSP assignment.

Fig. 6 shows the ROC curves for each SCOP class and for the full dataset (856 proteins, 143 202 residues). The global area under the curve (AUC) is 0.720, significantly above the random baseline of 0.5. The class dependence is weak: AUC ranges from 0.667 (all- β) to 0.739 (all- α), with α/β and $\alpha + \beta$ at 0.713 and 0.714 respectively. Even in all- β proteins, where helical residues constitute only 15% of the total (4741 out of 30 980), the AUC remains well above chance. This confirms that the integrability error operates at the residue level, detecting local geometric regularity rather than relying on the overall fold type.

The condition $E[n] < E_{\text{th}}$ tests whether the backbone at residue n preserves discrete helical symmetry, that is, whether the local curva-

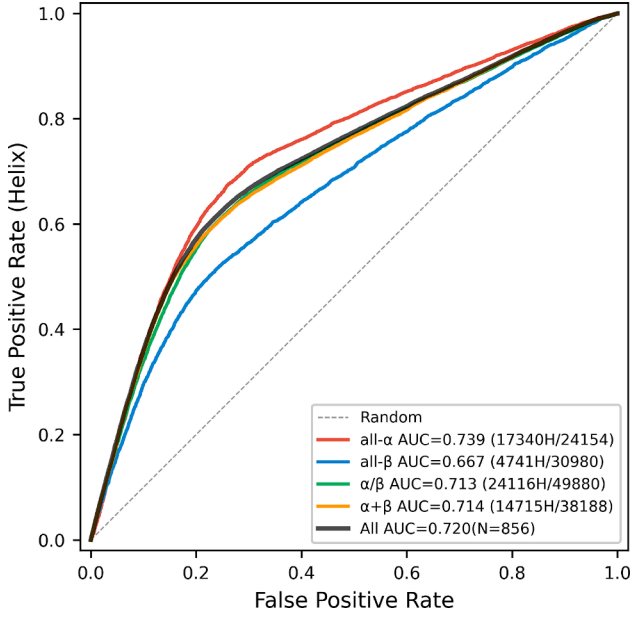


Fig. 6. ROC curve for helix detection using the integrability error $E[n] = |\cos \tau[n] - (1 + V_{re}[n]/2\beta)|$ as a binary classifier (helix vs. non-helix by DSSP). Curves are shown for each SCOP class and for the full dataset (856 proteins, 143 202 residues). Parentheses indicate the number of helix residues over total residues in each class. The global AUC = 0.720 means that a randomly chosen helical residue has lower $E[n]$ than a randomly chosen non-helical residue 72% of the time. The class dependence is weak (AUC range 0.667–0.739), confirming that the integrability–helicity correspondence operates at the residue level rather than depending on global fold type. The gap between AUC = 0.72 and unity quantifies the structural information carried by hydrogen bonds, non-local contacts, and torsion-sign degrees of freedom that lie outside the scalar dispersion relation.

ture and torsion are sufficiently uniform that a single screw motion $\mathbf{r}[n+1] = M \mathbf{r}[n]$ approximately generates the chain segment. When $\kappa[n]$ and $\tau[n]$ are exactly constant, the curvature ratios satisfy $r^\pm = 1$, the exact decomposition (Eq. (13)) reduces to the dispersion relation (Eq. (24)), and $E[n] \equiv 0$; the geometric origin of this vanishing is analyzed in Section 4.6, where we show that $E[n]$ is an order parameter for discrete helical symmetry breaking. In this sense, α -helices are the backbone regions where the DNLS most closely approximates an integrable system, and $E[n]$ measures the degree to which this integrability is locally broken.

4.5. The protein backbone as a piecewise integrable system

The results of this section lead to a unified geometric picture of protein backbone structure in the DNLS framework. The backbone can be viewed as a piecewise integrable system: α -helical segments are regions of approximate integrability where the curvature and torsion are locally uniform, the dispersion relation (Eq. (24)) holds, and the DNLS admits soliton-like solutions. β -strand and coil regions, by contrast, are regions of broken integrability where $r^\pm$ deviates significantly from unity, the dispersion relation fails, and the full decomposition (Eqs. (13)–(14)) is required.

This picture provides a geometric definition of secondary structure that is independent of hydrogen-bond criteria. An α -helix is a contiguous segment where the DNLS integrability error $E[n]$ remains below a threshold; a non-helical region is one where $E[n]$ exceeds this threshold. The definition is purely kinematic: it depends only on the C_α coordinates and requires no energy function or force field.

Helical segments exhibit the smallest dispersion-relation error among all secondary-structure types (Fig. 5), but they are not the sole regions of approximate integrability. Short segments within β -strands

or loops may also satisfy $r^\pm \approx 1$ locally, contributing to the imperfect separation in the ROC analysis (Fig. 6). The gap between AUC = 0.72 and unity reflects both these local exceptions and the information carried by V_{im} (torsion-angle signs) and non-local interactions (hydrogen bonds, tertiary contacts) that lie outside the scalar dispersion relation.

The dispersion relation has two limitations as a structural probe. First, Eq. (24) determines only $|\tau|$, not $\text{sign}(\tau)$. The torsion-sign ambiguity identified in Section 3.3 therefore persists even within integrable segments. Second, the dispersion relation provides no information about the spatial arrangement of secondary-structure elements relative to one another. It is a local diagnostic that characterizes individual residues but cannot address the global fold topology. These limitations reinforce the conclusion that the local, real-potential reduction is a geometric analysis tool rather than a predictive framework.

4.6. Geometric interpretation of the integrability error

The integrability error $E[n]$ defined in Eq. (26) admits a direct geometric interpretation as a measure of broken discrete helical symmetry. We make this connection explicit.

A discrete curve $\{\mathbf{r}[n]\}$ possesses *discrete helical symmetry* if there exists a rigid screw motion M (a rotation about a fixed axis composed with a translation along that axis) such that $\mathbf{r}[n+1] = M \mathbf{r}[n]$ for all n . This condition requires that the discrete Frenet curvature and torsion be site-independent:

$$\kappa[n] = \kappa_0, \quad \tau[n] = \tau_0, \quad \forall n. \quad (28)$$

Substituting Eq. (28) into the exact decomposition (Eq. (13)) with uniform coupling $\beta_n^+ = \beta_n^- = \beta$ gives $r^\pm[n] = 1$ and $\tau[n+1] = \tau[n] = \tau_0$, so that

$$V_{re} = \beta \cos \tau_0 + \beta \cos \tau_0 - 2\beta = 2\beta(\cos \tau_0 - 1), \quad (29)$$

which is precisely the dispersion relation (Eq. (24)). The integrability error therefore vanishes identically:

$$E[n] = \left| \cos \tau_0 - \left(1 + \frac{2\beta(\cos \tau_0 - 1)}{2\beta} \right) \right| = 0. \quad (30)$$

The uniform condition Eq. (28), with κ and τ each held constant from site to site, is precisely that of a discrete *circular* helix: the orbit of a single rigid screw motion, with constant radius and constant pitch. This differs from the weaker notion supplied by the classical theorem of Lancret [34,35], which states that a space curve is a *generalized* helix if and only if the ratio κ/τ is constant. A generalized helix permits κ and τ to vary in concert so long as their ratio is fixed; the circular helix requires each to be constant separately and is therefore strictly stronger. The vanishing of the integrability error selects the stronger object: $E[n] = 0$ is the algebraic signature of a discrete circular helix, not of a Lancret generalized helix. Because $E[n]$ enters only through $\cos \tau[n]$, it fixes the curvature κ and the torsion magnitude $|\tau|$, whereas the sign of τ , the handedness of the screw orbit, remains an independent datum, precisely the chiral degree of freedom isolated in Barrier I (Section 3.3).

This identification gives $E[n]$ a geometric interpretation beyond that of a fitting residual: it serves as a *local measure of discrete helical symmetry breaking*. At each residue, $E[n]$ quantifies the degree to which the local backbone geometry deviates from a perfect discrete circular helix, i.e., from a fixed point of the screw-motion group. In this picture, helical segments correspond to near-symmetric regions ($E[n] \approx 0$), while loops and strands correspond to regions of broken symmetry ($E[n] \gg 0$). The circular helix is the differential-geometric realization of the uniform integrable segment analyzed dynamically above: the screw-motion orbit $\mathbf{r}[n+1] = M \mathbf{r}[n]$ is the same object as the translation-invariant Bloch mode of the transfer matrix T , whose conserved quasi-momentum coincides in magnitude with the torsion, $|q| = |\tau|$ (Eq. (24)). The geometric condition $E[n] = 0$ and the dynamical condition of an exactly conserved quasi-momentum are thus two readings of one structure, so that $E[n]$ measures, equivalently, the local breaking of helical symmetry and the local non-conservation of the Bloch quasi-momentum.

4.7. Integrable rigidity and the descriptive–predictive boundary

The transfer-matrix reading of the uniform segment (Section 4, Eq. (25)) does more than locate helical segments: it fixes how much structural freedom such a segment retains. The argument rests on a single elementary observation, that a uniform segment is a constant-coefficient linear recurrence, and holds within the class of purely local, real-potential reductions for which the dispersion relation Eq. (24) was derived. We use the term *integrable segment* here in its weakest sense, a segment that is translation-invariant and carries one conserved quasi-momentum.

Lemma 2 (Uniform-segment recurrence and conserved quasi-momentum). *Let $S = \{n_0, \dots, n_1\}$ be a segment on which the curvature ratios satisfy $r^\pm[n] \approx 1$, the couplings are uniform $\beta_n^\pm \approx \beta$, and the effective potential is real and constant, $V_{\text{eff}}[n] \equiv V_{\text{re}}$. Then the DNLS Eq. (7) reduces on S to the constant-coefficient, second-order linear recurrence $\psi[n+1] = a\psi[n] - \psi[n-1]$ with $a = 2 + V_{\text{re}}/\beta$, governed by the transfer matrix T of Eq. (25) with $\det T = 1$ and Bloch quasi-momentum $|q| = |\tau|$ fixed by Eq. (24). Because T is independent of n , the site shift is a symmetry and q (equivalently the discrete Wronskian) is conserved along S .*

Proof. Setting $\beta_n^\pm = \beta$ and $V_{\text{eff}}[n] = V_{\text{re}}$ in Eq. (7) turns the left-hand side into β times the discrete second difference $\psi[n+1] - 2\psi[n] + \psi[n-1]$; equating to $V_{\text{re}}\psi[n]$ and dividing by $\beta > 0$ gives the stated recurrence, whose transfer matrix, eigenvalues $\lambda_\pm = e^{\pm iq}$, and dispersion $|q| = |\tau|$ are those established in Eqs. (24)–(25) above. Constancy of a makes T commute with the site shift, conserving q . $\square$

Here V_{re} is read from the known backbone geometry rather than solved for self-consistently; the lemma is a kinematic statement about the potential profile of a given segment, in keeping with the role of the Hasimoto map as a kinematic identity.

Proposition 2 (Boundary rigidity). *Under the hypotheses of Lemma 2, the solution space on S is exactly two-dimensional, and the two boundary data, namely the endpoint values, or equivalently a single adjacent pair $(\psi[n_0], \psi[n_0-1])$ (an initial position, tangent, and normal in the reconstruction of the lines preceding Eq. (4)), determine the entire interior $\{\psi[n] : n_0 < n < n_1\}$ uniquely. No interior degree of freedom remains.*

Proof. Since $\det T = 1 \neq 0$, T is invertible and the map from a solution to its seed $(\psi[n_0], \psi[n_0-1]) \in \mathbb{C}^2$ is a linear isomorphism: the seed determines $\psi[n] = (T^{n-n_0} \text{seed})_1$ for every n , forward and backward. Hence the solution space is two-dimensional and one adjacent-pair seed fixes the interior uniquely. (The two-endpoint form is equivalent away from the measure-zero standing-wave resonances $q(n_1 - n_0) \in \pi\mathbb{Z}$.) At the band edges $q \in \{0, \pi\}$ the roots coincide and the second solution is the secular mode $(A + Bn)(\pm 1)^n$; the space is still two-dimensional, so the conclusion is unchanged. $\square$

Corollary 2 (Description, not prediction). *Within the purely local, real-potential reduction, the interior conformation of an integrable segment is fixed by (β, V_{re}) together with the two boundary data (Proposition 2). The amino-acid sequence enters this interior only through the bond stiffnesses $\beta_n^\pm = 1/|t_n|$ within the single scalar V_{re} ; the dispersion relation $\cos \tau = 1 + V_{\text{re}}/2\beta$ carries no other sequence-dependent quantity, and this channel accounts for less than 5% of the variance of V_{re} on our dataset (Section 3), the remainder being set by the local geometry (κ, τ) . The conserved quasi-momentum $|q| = |\tau|$ is thus a quantity read off a known segment rather than generated from sequence: an integrable reduction describes a backbone whose endpoints are given, but does not predict an interior conformation from sequence. The map that de novo prediction would require, from sequence to a specific three-dimensional interior of dimension $2(L-1)$ for a segment of length L , is precisely what the boundary rigidity removes.*

Interpretation. A uniform segment is translation-invariant, carries a single conserved quasi-momentum, and is fixed by its boundaries; this rigidity is what makes the segment integrable and, at the same time,

what prevents its interior from encoding a sequence-dependent fold. The conclusion is a property of the local, real-potential reduction, not of the Hasimoto map, and it constrains only the interior of a segment. The relative placement of secondary-structure elements is a separate matter, governed by information the local dispersion relation does not carry (Section 4). Nor does it say that sequence lacks the physico-chemical information needed to fold a protein: that information is real, but the present model does not couple it into the quantities that fix the interior. A predictive model must do exactly that, incorporating amino-acid information into its parameters. This is consistent with the predictive successes of DNLS-based models, which supply the missing conformational information externally, through dissipative relaxation dynamics (Monte Carlo or Glauber, rather than integrable time evolution) together with injection of the native structure via fitted parameters, native-seeded external fields [11], or an RMSD-to-native term in the cost functional [4]. In a genuine *de novo* setting that injected information is absent, and the boundary data the rigidity requires is itself the missing answer.

5. Self-consistent field test

The two barriers identified in Section 3 are static: they concern the information content of V_{eff} extracted from known structures and are properties of the local, real-potential reduction itself. They follow from discarding V_{im} and from the geometric dominance of V_{re} (Eqs. (13)–(14)), and hence persist for any choice of energy-function parameterization within this model class. This point is demonstrated directly by what we term an *oracle test*: in Fig. 1(b) we supply the exact $V_{\text{re}}[n]$ computed from the native structure, which constitutes the best possible real potential that any energy function could produce, and attempt backbone reconstruction with $V_{\text{im}} = 0$. The resulting RMSD of 20–120 Å across all chain lengths (mean ~ 50 Å; 40–120 Å for the longest 200–300-residue chains, cf. Section 3) shows that even perfect knowledge of V_{re} is insufficient, because the 2^N torsion-sign degeneracy is intrinsic to the real-potential reduction and cannot be resolved by any choice of energy function.

A separate question is whether the DNLS can be used as a dynamical equation to drive an initially unfolded chain toward the native state, given a physically motivated effective potential. A natural dynamical hypothesis is that the DNLS could drive folding through modulation instability, in which small perturbations of a uniform solution grow exponentially and might seed the formation of soliton-like secondary-structure elements. We test this dynamical scenario directly by performing self-consistent field (SCF) iterations on all 856 proteins in our dataset, using two independent settings: one with hydrophobic and elastic potentials only, and one with an additional hydrogen-bond term. The SCF test provides a third, dynamical barrier that is independent of Barriers I and II and serves as an additional confirmation of the representational limitation.

5.1. SCF formulation

We initialize the complex field as a nearly uniform state $\psi_0[n] = \kappa_0 + \delta\psi[n]$, where $\kappa_0 = 0.1$ rad represents a nearly straight chain and $\delta\psi[n]$ is a small random perturbation with $|\delta\psi| < 0.01$. The field is evolved under damped dynamics

$$\frac{d\psi[n]}{dt} = -(1 + i\gamma) \frac{\delta H[\psi]}{\delta \psi^*[n]}, \quad (31)$$

where $\gamma = 0.5$ is a dissipation parameter that drives the system toward a local energy minimum rather than conserving the Hamiltonian. The functional $H[\psi]$ consists of a kinetic (elastic) term and an interaction term:

$$H[\psi] = \sum_n \beta |\psi[n+1] - \psi[n]|^2 + \sum_n V_{\text{int}}[n] |\psi[n]|^2. \quad (32)$$

The interaction potential V_{int} is constructed from two contributions. The elastic term $V_{\text{elastic}}[n] = \lambda(\kappa[n] - \kappa_{\text{target}})^2$ penalizes deviations of the

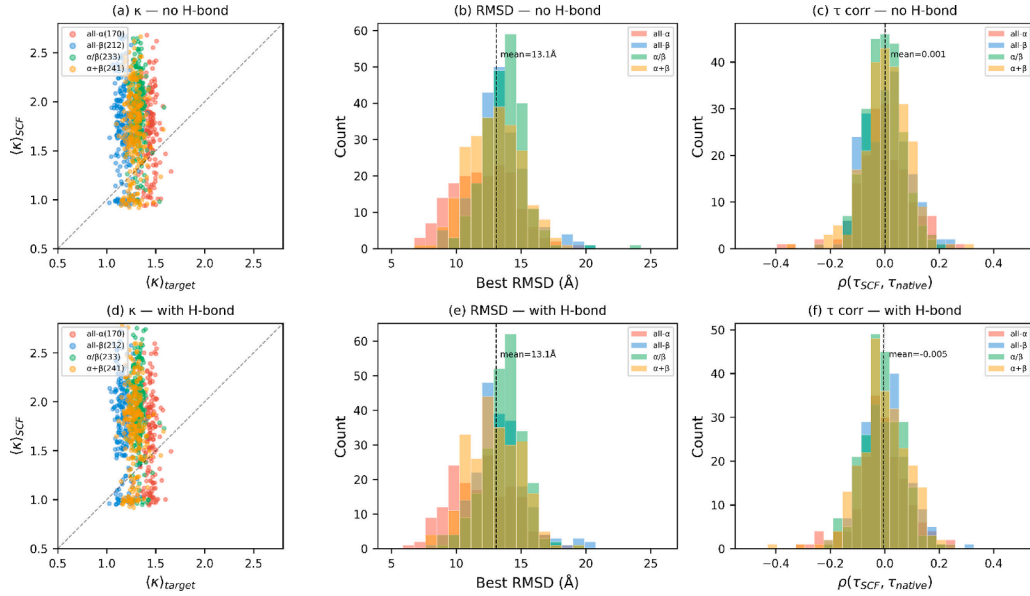


Fig. 7. Self-consistent field (SCF) test of DNLS-driven folding on 856 non-redundant proteins, without (top row) and with (bottom row) a hydrogen-bond potential. Points and histograms are colored by SCOP class: all- α (170, red), all- β (212, blue), α/β (233, green), $\alpha + \beta$ (241, orange). (a, d) Mean bond curvature $\langle \kappa \rangle_{SCF}$ vs. native target $\langle \kappa \rangle_{target}$; the dashed line marks perfect agreement. In both settings $\langle \kappa \rangle_{SCF}$ is systematically overestimated (mean 1.776 and 1.790 vs. target 1.298), with no SCOP class approaching the diagonal. (b, e) Distribution of best-achieved RMSD. The two histograms are statistically indistinguishable (mean 13.1 Å; range 6.8–24.0 Å without hydrogen bonds, 6.7–26.4 Å with hydrogen bonds), and no protein in either setting reaches RMSD < 5 Å. (c, f) Pearson correlation $\rho(\tau_{SCF}, \tau_{native})$ between the SCF-predicted and native torsion-angle profiles. Both distributions are centered on zero (mean 0.001 vs. –0.005), indicating that the SCF torsion angles are uncorrelated with the native structure. The near-identical results across the two rows demonstrate that the folding failure is not due to missing hydrogen-bond physics but to the real-potential SCF dynamics on the single complex scalar field ψ , whose dissipative evolution couples amplitude and phase and so cannot independently regulate the curvature and torsion constraints required for structure determination.

local curvature from a target value $\kappa_{target} = \langle \kappa \rangle_{native}$ computed from the known structure. The hydrophobic term $V_{hydro}[n]$ is a non-local potential that assigns a favorable energy to contacts between hydrophobic residues:

$$V_{hydro}[n] = - \sum_{m, |m-n| > 4} h[n] h[m] f(|\mathbf{r}[n] - \mathbf{r}[m]|), \quad (33)$$

where $h[n] \in \{0, 1\}$ is the hydrophobicity index of residue n (assigned according to the Kyte-Doolittle [36] scale with a binary cutoff) and $f(r)$ is a contact function that equals unity for $r < 8$ Å and decays smoothly to zero beyond this distance. The backbone coordinates $\mathbf{r}[n]$ are reconstructed from $\psi[n]$ at each SCF step via the inverse Hasimoto transform (Eq. (6)) and discrete Frenet reconstruction.

In the second setting we add a hydrogen-bond potential

$$V_{hb}[n] = -\epsilon_{hb} \sum_{m=n+3}^{n+5} g(\kappa[n], \kappa[m]) f(|\mathbf{r}[n] - \mathbf{r}[m]|), \quad (34)$$

where ϵ_{hb} is a coupling strength and $g(\kappa[n], \kappa[m])$ is a geometric filter that favors the curvature values characteristic of helical hydrogen bonds. This term is designed to mimic the $i \rightarrow i + 4$ backbone hydrogen bond that stabilizes α -helices, expressed entirely in terms of the ψ field variables.

The SCF iteration proceeds as follows. Starting from ψ_0 , we (i) reconstruct the backbone coordinates, (ii) evaluate V_{int} (and V_{hb} in the second setting), (iii) update ψ according to Eq. (31) with a discrete time step $\Delta t = 0.01$, and (iv) repeat for 5000 steps. At each step we record the mean curvature $\langle \kappa \rangle = N^{-1} \sum_n |\psi[n]|$ and the C_α RMSD between the reconstructed backbone and the native structure (after optimal superposition). The best RMSD achieved over the trajectory is reported as the outcome for each protein.

5.2. Results without hydrogen bonds

Fig. 7(a) shows the final mean curvature $\langle \kappa \rangle_{SCF}$ plotted against the native target $\langle \kappa \rangle_{target}$ for all 856 proteins, colored by SCOP class.

The dashed diagonal marks perfect agreement. The SCF systematically overestimates the mean curvature: the dataset-wide mean of $\langle \kappa \rangle_{SCF}$ is 1.776 rad compared to a target of 1.298 rad, an overestimation of approximately 37%. No SCOP class approaches the diagonal. The overestimation reflects the tendency of the damped DNLS dynamics to produce excessive bending: the hydrophobic potential drives chain compaction, which in the ψ representation manifests as increased $|\psi| = \kappa$, but without the directional constraints of hydrogen bonds there is no mechanism to regulate the curvature to its native value.

Fig. 7(b) displays the distribution of best-achieved RMSD across the 856 proteins. The mean RMSD is 13.1 Å with a range of 6.8 to 24.0 Å. No protein achieves RMSD below 5 Å, a threshold commonly used to indicate a successful fold prediction. The distribution shows no significant separation among the four SCOP classes: all- α , all- β , α/β , and $\alpha + \beta$ proteins are interleaved throughout the histogram. This uniformity confirms that the SCF failure is not specific to any particular fold topology.

Fig. 7(c) shows the Pearson correlation $\rho(\tau_{SCF}, \tau_{native})$ between the torsion-angle profile produced by the SCF and the native torsion-angle profile, computed for each protein. The distribution is centered on zero (mean $\rho = 0.001$, median $\rho = 0.000$) and extends symmetrically over the range $[-0.4, 0.4]$. The SCF-generated torsion angles therefore bear no systematic relationship to the native structure.

The absence of correlation between predicted and native torsion angles constitutes a clear failure of the *de novo* predictive scheme. Although the SCF dynamics may induce chain compaction, reflected in the curvature profile, the torsion angles $\tau[n]$, which govern the global fold topology, remain effectively stochastic. The result is a compact globule with no native structural fidelity: the isotropic hydrophobic potential drives compaction (increased κ) but cannot select the specific torsion-angle sequence that defines the native fold. This is consistent with the established distinction between hydrophobic collapse and folding. The selection of specific secondary and tertiary structure elements involves directional hydrogen bonds, whose geometric constraints are difficult to represent within the scalar DNLS framework.

5.3. Results with hydrogen bonds

The bottom row of Fig. 7 shows the same three diagnostics for the SCF with the hydrogen-bond potential (Eq. (34)) included. The results are statistically indistinguishable from the setting without hydrogen bonds. The mean curvature increases slightly to $\langle \kappa \rangle_{\text{SCF}} = 1.790$ rad (compared to 1.776 without the hydrogen-bond term), moving further from the target rather than closer. The RMSD distribution is virtually identical: mean 13.1 Å, range 6.7 to 26.4 Å, with no protein reaching RMSD < 5 Å. The torsion correlation shifts negligibly from $\rho = 0.001$ to $\rho = -0.005$, remaining consistent with zero.

The universal failure of the hydrogen-bond term to improve the SCF outcome demonstrates that the barrier to folding within the DNLS framework is not merely energetic, but representational: the single complex scalar field $\psi[n] = \kappa[n] \exp(i \sum \tau[k])$ entangles curvature and torsion in its amplitude and phase. A physical backbone hydrogen bond imposes strictly independent constraints on spatial distance (governed by κ) and directional orientation (governed by τ) between specific residue pairs. The evolution of ψ couples its amplitude $|\psi|$ and phase $\arg(\psi)$ through the nonlinear dynamics, which prevents the independent regulation of curvature and torsion required to stabilize specific donor-acceptor geometries. The hydrogen-bond potential (Eq. (34)), expressed in terms of ψ , therefore cannot enforce the discrete orientational constraints that the physical interaction requires.

A natural question is whether replacing the interaction potential V_{int} with a more refined functional, for instance a C_α -level effective potential obtained by systematic coarse-graining of an all-atom force field such as AMBER [37,38] or CHARMM [39,40], would alter this conclusion. Three independent lines of evidence indicate that it would not. First, the oracle test Fig. 1(b) already supplies the exact native $V_{\text{re}}[n]$, which represents the theoretical optimum for the real part of the potential; the resulting RMSD of 20–120 Å demonstrates that the 2^N torsion-sign degeneracy (Barrier I) persists irrespective of energy-function quality. Second, the geometric dominance result (Barrier II, $\rho_{\text{geom}} = 0.951$) establishes that V_{re} is 95% determined by backbone geometry rather than by amino-acid identity, so that even an exact force-field projection would produce a V_{re} profile nearly indistinguishable from the geometry-only version. Third, the representational bottleneck is structural rather than parametric: as described above, the Hasimoto field ψ encodes the two real degrees of freedom κ and τ in a single complex number whose amplitude and phase remain coupled under the dynamics, so that no force field, however accurately parameterized, can independently enforce the distance and angular constraints that directional interactions impose on specific residue pairs.

The SCF test with its deliberately minimal energy function therefore probes the representational ceiling of the ψ framework, not the quality of a particular force field.

This conclusion extends to state-of-the-art machine-learning force fields (MLFFs) such as AI2BMD [41], MACE [42], and ViSNet [43]. These architectures achieve near-quantum accuracy by operating on explicit Cartesian coordinates, but the three arguments above apply to them unchanged: the bottleneck is representational, not energetic, so higher chemical fidelity cannot overcome the limitation of the Hasimoto framework. Because V_{re} is determined $\sim 95\%$ by geometry rather than sequence (Barrier II), the high-fidelity chemical information provided by MLFFs is largely filtered out when projected onto the DNLS potential. The real-potential reduction, which retains only V_{re} and discards the parity-odd V_{im} , thus acts as a lossy projection that drops the chiral degrees of freedom needed to distinguish the native state, a structural deficit that no improvement in force-field accuracy can overcome. This same limitation is recognized empirically in the MLFF literature: recent work reports that most existing potentials assume localized interactions and that explicitly modeling *non-local* electrostatic and multipole contributions reduces energy and force errors by more than 50% in biomolecular simulations [44]. Our parity-locality classification (Table 2) offers a first-principles account of why such non-local terms are

indispensable: the interactions that select tertiary structure populate the non-local row that any nearest-neighbor scalar potential, including the DNLS, cannot represent.

5.4. Summary of the dynamical barrier

The SCF experiments establish a third, dynamical barrier to using the local, real-potential DNLS reduction for protein structure prediction. The barrier can be stated concisely: the dissipative DNLS dynamics, even supplemented with physically motivated interaction terms, cannot drive an unfolded chain to the native state, because the single complex scalar field ψ lacks the representational capacity to encode the independent curvature and torsion constraints imposed by directional hydrogen bonds. Three quantitative findings support this conclusion across all 856 proteins and both SCF settings. First, the mean curvature is systematically overestimated by approximately 37%, indicating that the dynamics produces excessive bending without the regulatory effect of hydrogen-bond geometry. Second, no protein achieves RMSD below 5 Å, and the RMSD distribution shows no dependence on SCOP class, confirming that the failure is universal rather than fold-specific. Third, the torsion correlation with the native structure is indistinguishable from zero ($\rho \approx 0$), demonstrating that the SCF dynamics generates effectively random torsion-angle sequences. Comparing the two SCF settings isolates the cause of the failure. The addition of a hydrogen-bond potential within the DNLS framework produces no measurable improvement in any of the three diagnostics. This rules out the interpretation that the SCF failure is due to an incomplete energy function. The failure is instead structural, originating in the κ - τ entanglement of ψ described above. This *de novo* failure does not contradict the established predictive successes of DNLS-based models: when the free-energy parameters are fit to a known native structure, Monte Carlo (or Glauber) relaxation can indeed recover that structure from a random-walk phase [9,11]. Our claim is restricted to the *de novo* setting, in which no native structure is available to be injected, whether into the parameters, an external field, or an RMSD-to-native cost, and the dissipative dynamics is driven only by sequence-level potentials. The relationship between this dynamical barrier and the two static barriers of Section 3 is discussed in Section 6.

6. Discussion and conclusion

The Hasimoto transform was originally constructed for vortex filaments evolving under the local induction approximation (LIA). In that setting the transform is both kinematic and dynamic: it converts the geometric evolution of the filament into the integrable cubic NLS, whose soliton solutions describe physical excitations that propagate without dispersion. The success of this construction rests on four properties of the vortex system that proteins do not share: locality of interactions, homogeneity of the medium, Hamiltonian (non-dissipative) dynamics, and a physically determined effective potential. Protein folding violates all four. Hydrophobic contacts and electrostatic forces are non-local; the backbone comprises 20 chemically distinct amino acids; folding is a dissipative free-energy minimization in aqueous solvent; and, as shown in Section 3, the effective potential V_{eff} on proteins is an algebraic consequence of the backbone geometry rather than an independently specified physical quantity. Table 3 summarizes these distinctions. Each of the four mismatches removes a degree of freedom that the predictive problem requires: non-locality, chemical heterogeneity, dissipation, and a sequence-specified potential are precisely the ingredients that a local, homogeneous, Hamiltonian, geometry-fixed reduction leaves out. The reduction is in this sense kinematically complete but dynamically inert: it faithfully re-expresses a given geometry, yet retains none of the sequence-specific, non-local information from which that geometry would have to be generated.

The geometric formalism established by Niemi and collaborators provides a rigorous basis for the structural characterization of protein backbones. The identification of discrete soliton solutions with

Table 3

Comparison of the Hasimoto map applied to vortex filaments and to protein backbones. The four properties listed are necessary for the map to function as a predictive dynamical framework.

Property	Vortex filament	Protein backbone
Interactions	Local (LIA)	Non-local
Medium	Homogeneous	Heterogeneous (20 AA)
Dynamics	Hamiltonian	Dissipative
V_{eff}	Physical potential	Kinematic identity

secondary-structure motifs reveals the geometric regularity underlying secondary structure, while multi-soliton ansätze yield compact parameterizations of folded states with sub-ångström accuracy [7,8]. Recent extensions have successfully applied this basis to simulate thermal unfolding [9] and characterize topological phase transitions [10]. These studies constitute a comprehensive treatment of the *inverse problem*: given a known topology, the DNLS soliton basis affords an efficient representation and a robust framework for analyzing fluctuations around the native state. The topological aspects of this program connect to a broader body of work on protein topology. Approximately 6% of known protein structures form knots, slipknots, or links whose folding requires the backbone to cross topological barriers [18,19]. These entangled proteins highlight a fundamental limitation of any local geometric description: the topological invariants that distinguish a knotted from an unknotted fold are inherently global properties that cannot be fully determined from the local fields ($\kappa[n], \tau[n]$) at any finite number of sites. Consequently, soliton-based studies typically determine parameters by fitting to a known crystallographic structure, whether through Metropolis minimization of RMSD [7] or direct fitting to native coordinates [9].

That a known fold can be represented so faithfully has a precise structural explanation in the integrability hierarchy of discrete nonlinear equations. The geometric free energies employed in this program are, after the Hasimoto map, *on-site* discrete nonlinear Schrödinger functionals: their nonlinear terms (e.g. the quartic double well $\lambda(\kappa_i^2 - m^2)^2$ and the coupling $\kappa_i^2 \tau_i^2$) are diagonal in the site index, the only inter-site term being the linear hopping $\kappa_{i+1} \kappa_i$ [10]. Unlike the Ablowitz–Ladik lattice [45], the integrable discretization of NLS, which carries a Lax pair and an infinite hierarchy of conserved quantities, the on-site discrete NLS is non-integrable: it possesses no Lax pair and no conserved hierarchy beyond the global phase, and its uniform-segment limit retains only the single quasi-momentum of the constant-coefficient transfer matrix (25) (the discrete probability current of Eq. (22) is the conserved bilinear of the *linear* eigenproblem, distinct from the integrable *hierarchy* absent here). The α -helices are therefore isolated segments of *piecewise linear integrability*, not manifestations of a globally integrable dynamics. This is the structural origin of the descriptive–predictive asymmetry: a fully integrable system is rigidly constrained by its conserved hierarchy, so initial local data propagate to a unique global trajectory; the on-site model has no such rigidity, and its soliton solutions are obtained by direct energy minimization against a known target rather than by integrating the sequence forward. This structural feature also motivates the approach we take throughout. Rather than following the usual integrable-systems route of solving the equation for explicit soliton solutions and examining their stability, we take the integrable state itself as a fixed kinematic reference and measure how real backbones depart from it, using the exact decomposition, the integrability error $E[n]$, and the parity classification. Because the protein potential is reconstructed from a known geometry, solving for solitons would merely return the structure already supplied; the productive direction is the reverse one, reading the integrable ideal as a ruler for the deviations that carry the folding information. The absence of a Lax pair is thus not a technical detail but the reason the framework can represent a known fold yet cannot generate an unknown one from local sequence data alone.

In contrast, the present work interrogates the complementary *forward problem*: determining whether the DNLS framework permits *ab*

initio prediction of the native structure solely from the amino-acid sequence. Our analysis indicates that structural barriers impede this predictive pathway. The exact decomposition (Section 3) reveals that the effective potential V_{eff} is determined predominantly by the target geometry rather than the sequence, creating an informational circularity that cannot be resolved by sequence-based potentials. The SCF experiments (Section 5) further show that physically motivated interactions fail to lift the 2^N torsion-sign degeneracy or overcome the representational limitations of the scalar field. These findings indicate that the distinction between the inverse and forward problems is structural: the capacity to describe known folds does not imply the feasibility of predicting unknown ones. The specificity that prediction demands is precisely the non-local, sequence-dependent content discarded in passing to a local real-potential model, and rearranging that model does not restore it. One might nonetheless hope to supply that missing specificity dynamically. The hypothesis that modulation instability of the DNLS may provide a dynamical mechanism for the emergence of secondary structure from an initially straight chain is directly tested by our SCF experiments. The instability does produce curvature growth from a nearly uniform initial state, consistent with the expected behavior. However, the resulting structures bear no resemblance to native folds: the torsion angles are uncorrelated with the native profile ($\rho \approx 0$), and the RMSD remains above 5 Å for all 856 proteins in both SCF settings. Modulation instability generates generic bending but cannot select the specific (κ, τ) sequence that defines a particular protein fold.

Although this on-site, real-potential reduction cannot serve as a predictive tool, the exact decomposition and the analyses built upon it yield three constructive results. First, the integrability error $E[n]$ (Eq. (26)) serves as a geometric probe of secondary structure. The ROC analysis (Fig. 6, AUC = 0.72) demonstrates that 72% of the helix/non-helix distinction defined by DSSP is captured by a scalar test of discrete helical symmetry applied to C_α coordinates alone, without reference to hydrogen-bond energies or side-chain identities. Whether this geometric criterion can provide useful secondary-structure annotation in data-limited settings, such as low-resolution density maps or coarse-grained trajectories, remains to be tested on independent benchmarks; the present AUC establishes the conceptual correspondence between DNLS integrability and helical geometry but does not by itself validate a practical tool. The gap between AUC = 0.72 and unity quantifies the structural information that hydrogen bonds and non-local interactions contribute beyond local geometric regularity. Second, the effective potential $V_{\text{eff}}[n]$ provides a structural fingerprint that is invariant under rigid-body transformations. Because V_{re} is 95% determined by geometry (Section 3.4), homologous proteins with low sequence identity but similar folds produce similar $V_{\text{re}}[n]$ profiles, offering a one-dimensional scalar representation of backbone shape for structure comparison without spatial superposition. Third, the information-theoretic lower bound on chiral information loss is exact within the purely local, real-potential model class: each residue contributes one bit of torsion-sign information encoded in V_{im} but absent from V_{re} , producing a 2^N -fold degeneracy that any predictive scheme based on this class of models would need to resolve.

Comparison with the data-driven methods that have solved the prediction problem in practice throws these structural barriers into sharper relief. AlphaFold 2 [20] and ESMFold [23] both predict the full $SE(3)$ rigid-body frame, comprising a rotation matrix and a translation vector, for every residue. This approach retains the complete orientational degrees of freedom of the discrete Frenet frame without projecting them onto a scalar field. The theoretical foundations of this representational choice are clarified by the geometric deep learning framework [46,47], which demonstrates that neural architectures respecting the symmetry group of the data domain (specifically, $SE(3)$ equivariance for molecular geometry) achieve systematic gains in sample efficiency and generalization by building physical invariances directly into the network structure rather than learning them from data. The full-frame representation retains both signs of the torsion explicitly, whereas the local, real-

potential reduction of the DNLS, which keeps only $V_{\text{re}} = \text{Re } V_{\text{eff}}$, projects onto the parity-even sector, and our Barrier I shows that this projection discards the torsion sign, introducing a 2^N chiral degeneracy that the full-frame representation avoids entirely. Barrier II reveals a second contrast: the DNLS effective potential is 95% determined by local backbone geometry, whereas the attention mechanism of AlphaFold and the language-model embeddings of ESMFold encode long-range coevolutionary and contextual information that couples distant residues, information that statistical-physics methods such as direct coupling analysis [16,17] have shown to be extractable from evolutionary data via the maximum-entropy principle. The DNLS framework, operating on nearest-neighbor differences of ψ , has no analogous channel for non-local sequence information. These comparisons do not diminish the value of the geometric approach, since data-driven models supply predictions rather than physical explanations. They do, however, delineate the specific representational deficits that any future analytical theory must overcome: it must preserve the full $SE(3)$ frame at each residue and incorporate non-local, sequence-dependent interactions that go beyond the scalar Hasimoto field.

The three barriers identified in this work point to specific physical ingredients that a geometric theory of protein folding would need to incorporate. Hydrogen bonds impose simultaneous constraints on the distance and relative orientation of donor and acceptor groups, coupling κ and τ at specific residue pairs in a manner that cannot be captured by a potential acting on the single complex field ψ . A geometric energy functional treating $\kappa[n]$ and $\tau[n]$ as independent fields, rather than combining them into a single complex scalar, would be a natural starting point. The hydrophobic effect, an entropic force mediated by solvent reorganization, is inherently non-local and temperature-dependent, with no natural representation in the DNLS Hamiltonian. The SCF experiments confirm that a contact-based hydrophobic potential produces chain compaction but cannot select secondary structure, consistent with the known distinction between hydrophobic collapse and folding. Finally, the geometric dominance of V_{re} means that sequence-dependent terms going beyond the weak modulation of virtual-bond lengths in $\beta_n^\pm$ would need to enter any predictive geometric framework explicitly.

In summary, we have derived an exact closed-form decomposition of the DNLS effective potential on protein backbones and used it to identify three structural barriers to forward structure prediction: the torsion-sign degeneracy encoded in V_{im} , the geometric dominance of V_{re} , and the universal failure of self-consistent field dynamics across 856 non-redundant proteins. We further showed, in dynamical-systems terms, that the rigidity of an integrable uniform segment is what removes the freedom prediction requires, since boundary data fix the interior. A parity-by-locality classification of the folding forces then explains why the tertiary-structure drivers lie outside the local, real-potential sector. These barriers are mathematical in nature and are unlikely to be resolved by parameter adjustment or algorithmic improvement within the local, real-potential DNLS reduction. The Hasimoto map applied to proteins functions primarily as a kinematic identity rather than a dynamical equation: it provides a useful geometric language for describing folded states, a purely geometric helix detector, and a rotation-invariant structural fingerprint, but within the scope of our analysis the local, real-potential reduction built on it does not provide a viable pathway to *ab initio* protein structure prediction. As the comparison with data-driven methods above makes concrete, a future analytical theory of folding must go beyond the scalar Hasimoto field to preserve the geometric and chemical degrees of freedom that the folding process requires. The piecewise picture that emerges here, in which the integrability error partitions the backbone into near-integrable segments where $E[n] \rightarrow 0$ and the broken-integrability regions between them, suggests a natural question for future work: whether the boundary between the ordered, soliton-like helical phase and the disordered coil phase admits a sharp theoretical characterization, in the sense of a limiting form for the order parameter across the transition. The descriptive–predictive tension we locate here for the protein backbone may, more gener-

ally, be characteristic of integrable reductions of heterogeneous physical systems.

CRedit authorship contribution statement

Yiquan Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The source code for this study is available on GitHub at https://github.com/wyqmath/discrete_hasimoto_protein.

References

- [1] H. Hasimoto, A soliton on a vortex filament, *J. Fluid Mech.* 51 (3) (1972) 477–485.
- [2] C.B. Anfinsen, Principles that govern the folding of protein chains, *Science* 181 (4096) (1973) 223–230.
- [3] U.H. Danielsson, M. Lundgren, A.J. Niemi, Gauge field theory of chirally folded homopolymers with applications to folded proteins, *Phys. Rev. E-Stat. Nonlinear Soft Matter Phys.* 82 (2) (2010) 021910.
- [4] M. Chernodub, S. Hu, A.J. Niemi, Topological solitons and folded proteins, *Phys. Rev. E Stat. Nonlinear Soft Matter Phys.* 82 (1) (2010) 011916.
- [5] A. Molochkov, A. Begun, A. Niemi, Gauge symmetries and structure of proteins, in: *EPJ Web of Conferences*, 137, EDP Sciences, 2017, p. 04004.
- [6] D. Melnikov, A.B.F. Neves, Chern-Simons-Higgs model as a theory of protein molecules, *J. Appl. Phys.* 126 (24) (2019).
- [7] N. Molkenhuth, S. Hu, A.J. Niemi, Discrete nonlinear Schrödinger equation and polygonal solitons with applications to collapsed proteins, *Phys. Rev. Lett.* 106 (7) (2011) 078102.
- [8] A. Krokhotin, A.J. Niemi, X. Peng, Soliton concepts and protein structure, *Phys. Rev. E-Stat. Nonlinear Soft Matter Phys.* 85 (3) (2012) 031906.
- [9] A. Begun, S. Liubimov, A. Molochkov, A.J. Niemi, On topology and knotty entanglement in protein folding, *PLoS One* 16 (1) (2021) e0244547.
- [10] A. Begun, M.N. Chernodub, A. Molochkov, A.J. Niemi, Local topology and perestroikas in protein structure and folding dynamics, *Phys. Rev. E* 111 (2) (2025) 024406.
- [11] S.D. Liubimov, N.V. Gerasimeniuk, A.A. Korneev, A.V. Molochkov, Modeling the structure of myoglobin within the Abelian Higgs model, *Biochem. Suppl. A Membr. Cell Biol.* 19 (4) (2025) 449–455.
- [12] K.A. Dill, J.L. MacCallum, The protein-folding problem, 50 years on, *Science* 338 (6110) (2012) 1042–1046.
- [13] A. Davtyan, N.P. Schafer, W. Zheng, C. Clementi, P.G. Wolynes, G.A. Papoian, AWSEM-MD: protein structure prediction using coarse-grained physical potentials and bioinformatically based local structure biasing, *J. Phys. Chem. B* 116 (29) (2012) 8494–8503.
- [14] J.R. Banavar, A. Maritan, Physics of proteins, *Annu. Rev. Biophys. Biomol. Struct.* 36 (1) (2007) 261–280.
- [15] J.R. Banavar, A. Maritan, C. Micheletti, A. Trovato, Geometry and physics of proteins, *Proteins Struct. Funct. Bioinform.* 47 (3) (2002) 315–322.
- [16] F. Morcos, A. Pagnani, B. Lunt, A. Bertolino, D.S. Marks, C. Sander, R. Zecchina, J.N. Onuchic, T. Hwa, M. Weigt, Direct-coupling analysis of residue coevolution captures native contacts across many protein families, *Proc. Natl. Acad. Sci.* 108 (49) (2011) E1293–E1301.
- [17] S. Cocco, R. Monasson, M. Weigt, From principal component to direct coupling analysis of coevolution in proteins: low-eigenvalue modes are needed for structure prediction, *PLoS Comput. Biol.* 9 (8) (2013) e1003176.
- [18] J.I. Sulkowska, On folding of entangled proteins: knots, lassos, links and θ -curves, *Curr. Opin. Struct. Biol.* 60 (2020) 131–141.
- [19] P. Dabrowski-Tumanski, J.I. Sulkowska, Topological knots and links in proteins, *Proc. Natl. Acad. Sci.* 114 (13) (2017) 3415–3420.
- [20] J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Židek, A. Potapenko, et al., Highly accurate protein structure prediction with AlphaFold, *Nature* 596 (7873) (2021) 583–589.

- [21] J. Abramson, J. Adler, J. Dunger, R. Evans, T. Green, A. Pritzel, O. Ronneberger, L. Willmore, A.J. Ballard, J. Bambrick, et al., Accurate structure prediction of biomolecular interactions with AlphaFold 3, *Nature* 630 (8016) (2024) 493–500.
- [22] A. Rives, J. Meier, T. Sercu, S. Goyal, Z. Lin, J. Liu, D. Guo, M. Ott, C.L. Zitnick, J. Ma, et al., Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences, *Proc. Natl. Acad. Sci.* 118 (15) (2021) e2016239118.
- [23] Z. Lin, H. Akin, R. Rao, B. Hie, Z. Zhu, W. Lu, N. Smetanin, R. Verkuil, O. Kabeli, Y. Shmueli, et al., Evolutionary-scale prediction of atomic-level protein structure with a language model, *Science* 379 (6637) (2023) 1123–1130.
- [24] Y. Wang, M. Cai, Y. Ma, K. Wei, Sequence-context-aware decoding enables robust reconstruction of protein dynamics from crystallographic B-factors, (2025). *bioRxiv preprint*, <https://doi.org/10.64898/2025.12.05.692009>
- [25] R.A. Engh, R. Huber, Accurate bond and angle parameters for X-ray protein structure refinement, *Found. Crystallogr.* 47 (4) (1991) 392–400.
- [26] S.C. Lovell, I.W. Davis, W.B. Arendall, III, P.I.W. De Bakker, J.M. Word, M.G. Prisant, J.S. Richardson, D.C. Richardson, Structure validation by α geometry: ϕ , ψ and $C\beta$ deviation, *Proteins Struct. Funct. Bioinform.* 50 (3) (2003) 437–450.
- [27] J.C. Eilbeck, P.S. Lomdahl, A.C. Scott, The discrete self-trapping equation, *Phys. D Nonlinear Phenom.* 16 (3) (1985) 318–338.
- [28] P.G. Kevrekidis, *The Discrete Nonlinear Schrödinger Equation: Mathematical Analysis, Numerical Computations and Physical Perspectives*, vol. 232, Springer Science & Business Media, 2009.
- [29] A.G. Murzin, S.E. Brenner, T. Hubbard, C. Chothia, SCOP: a structural classification of proteins database for the investigation of sequences and structures, *J. Mol. Biol.* 247 (4) (1995) 536–540.
- [30] A. Andreeva, E. Kulesha, J. Gough, A.G. Murzin, The SCOP database in 2020: expanded classification of representative family and superfamily domains of known protein structures, *Nucleic Acids Res.* 48 (D1) (2020) D376–D382.
- [31] G. Wang, R.L. Dunbrack, Jr, PISCES: a protein sequence culling server, *Bioinformatics* 19 (12) (2003) 1589–1591.
- [32] Y. Wang, M. Cai, Y. Dong, Y. Ma, K. Wei, From signal to symphony: exploring 2D sequence representations for protein function prediction, *J. Chem. Inf. Model.* 65 (23) (2025) 12723–12736.
- [33] W. Kabsch, C. Sander, Dictionary of protein secondary structure: pattern recognition of hydrogen-bonded and geometrical features, *Biopolym. Orig. Res. Biomol.* 22 (12) (1983) 2577–2637.
- [34] M.-A. Lancré, *Mémoire sur les courbes à double courbure*, *Mem. Présentées à l'Institut* 1 (1806) 416–454.
- [35] M.P. do Carmo, *Differential Geometry of Curves and Surfaces*, Englewood Cliffs, New Jersey (1976).
- [36] J. Kyte, R.F. Doolittle, A simple method for displaying the hydropathic character of a protein, *J. Mol. Biol.* 157 (1) (1982) 105–132.
- [37] W.D. Cornell, P. Cieplak, C.I. Bayly, I.R. Gould, K.M. Merz, D.M. Ferguson, D.C. Spellmeyer, T. Fox, J.W. Caldwell, P.A. Kollman, A second generation force field for the simulation of proteins, nucleic acids, and organic molecules, *J. Am. Chem. Soc.* 117 (19) (1995) 5179–5197.
- [38] D.A. Case, T.E. Cheatham, III, T. Darden, H. Gohlke, R. Luo, K.M. Merz, Jr, A. Onufriev, C. Simmerling, B. Wang, R.J. Woods, The Amber biomolecular simulation programs, *J. Comput. Chem.* 26 (16) (2005) 1668–1688.
- [39] B.R. Brooks, R.E. Bruccoleri, B.D. Olafson, D.J. States, S.a. Swaminathan, M. Karplus, CHARMM: a program for macromolecular energy, minimization, and dynamics calculations, *J. Comput. Chem.* 4 (2) (1983) 187–217.
- [40] B.R. Brooks, C.L. Brooks, III, A.D. Mackerell, Jr, L. Nilsson, R.J. Petrella, B. Roux, Y. Won, G. Archontis, C. Bartels, S. Boresch, et al., CHARMM: the biomolecular simulation program, *J. Comput. Chem.* 30 (10) (2009) 1545–1614.
- [41] T. Wang, X. He, M. Li, Y. Li, R. Bi, Y. Wang, C. Cheng, X. Shen, J. Meng, H. Zhang, et al., Ab initio characterization of protein molecular dynamics with AI2BMD, *Nature* 635 (8040) (2024) 1019–1027.
- [42] I. Batatia, D.P. Kovacs, G. Simm, C. Ortner, G. Csányi, MACE: higher order equivariant message passing neural networks for fast and accurate force fields, *Adv. Neural Inf. Process. Syst.* 35 (2022) 11423–11436.
- [43] Y. Wang, T. Wang, S. Li, X. He, M. Li, Z. Wang, N. Zheng, B. Shao, T.-Y. Liu, Enhancing geometric representations for molecules with equivariant vector-scalar interactive message passing, *Nat. Commun.* 15 (1) (2024) 313.
- [44] T. Cui, Z. Wang, T. Wang, Enhancing non-local interaction modeling for ab initio biomolecular calculations and simulations with ViSNet-PIMA, (2026). *bioRxiv preprint*, <https://doi.org/10.64898/2026.03.18.712561>
- [45] M.J. Ablowitz, J.F. Ladik, Nonlinear differential–difference equations and Fourier analysis, *J. Math. Phys.* 17 (6) (1976) 1011–1018.
- [46] M.M. Bronstein, J. Bruna, T. Cohen, P. Veličković, Geometric deep learning: grids, groups, graphs, geodesics, and gauges, (2021). *arXiv preprint arXiv:2104.13478*
- [47] M.M. Bronstein, J. Bruna, Y. LeCun, A. Szlam, P. Vandergheynst, Geometric deep learning: going beyond euclidean data, *IEEE Signal Process. Mag.* 34 (4) (2017) 18–42.